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Solid-state imaging element and video recording apparatus

Abstract

A solid-state imaging element according to the present disclosure includes: a first semiconductor substrate that includes a floating diffusion that temporarily holds an electric signal output from a photoelectric conversion element; and a second semiconductor substrate that faces the first semiconductor substrate, in which the second semiconductor substrate includes a first transistor disposed on a side facing the first semiconductor substrate, the first transistor including: a channel extending along a thickness direction of the second semiconductor substrate; and a multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the channel, and the multi-gate of the first transistor is connected to the floating diffusion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/298,875, filed Jun. 1, 2021, which is a national stage application under 35 U.S.C. 371 and claims the benefit of PCT Application No. PCT/JP2019/044659, having an international filing date of Nov. 14, 2019, which designated the United States, which PCT application claimed the benefit of Japanese Patent Application No. 2018-233437 filed Dec. 13, 2018, the entire disclosures of each of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a solid-state imaging element and a video recording apparatus.

BACKGROUND ART

(2) There is a three-dimensional packaging technology for stacking a plurality of semiconductor substrates. For example, a solid-state imaging element having a known configuration includes a first semiconductor substrate on which a pixel region is formed and a second semiconductor substrate on which a logic circuit is formed (see Patent Document 1, for example).

CITATION LIST

Patent Document

(3) Patent Document 1: Japanese Patent Application Laid-Open No. 2010-245506

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(4) The solid-state imaging element disclosed in Patent Document 1 fails to secure a sufficient

space for a pixel transistor to be disposed. Then, a conceivable solution may be, for example, subdividing a substrate into a substrate on which a photoelectric conversion element is formed and a substrate on which a pixel transistor is formed, and stacking these substrates.

(5) However, such a configuration may need a longer wiring that connects the substrate on which the photoelectric conversion element is formed and the substrate on which the pixel transistor is formed, resulting in a reduced photoelectric conversion efficiency of the photoelectric conversion element.

(6) Therefore, the present disclosure proposes a solid-state imaging element and a video recording apparatus that are capable of improving the photoelectric conversion efficiency of the photoelectric conversion element by reducing the length of the wiring between stacked substrates.

Solutions to Problems

(7) A solid-state imaging element according to the present disclosure includes: a first semiconductor substrate that includes a floating diffusion that temporarily holds an electric signal output from a photoelectric conversion element; and a second semiconductor substrate that faces the first semiconductor substrate, in which the second semiconductor substrate includes a first transistor disposed on a side facing the first semiconductor substrate, the first transistor including: a channel extending along a thickness direction of the second semiconductor substrate; and a multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the channel, and the multi-gate of the first transistor is connected to the floating diffusion.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating an example schematic configuration of a solid-state imaging element applied to embodiments of the present disclosure.

(2) FIG. 2 is a diagram illustrating an example of the sensor pixel and readout circuit in FIG. 1.

(3) FIG. 3 is a diagram illustrating an example of the sensor pixel and readout circuit in FIG. 1.

(4) FIG. 4 is a diagram illustrating an example of the sensor pixel and readout circuit in FIG. 1.

(5) FIG. 5 is a diagram illustrating an example of the sensor pixel and readout circuit in FIG. 1.

(6) FIG. 6 is a diagram illustrating an example of a connection mode between a plurality of readout circuits and a plurality of vertical signal lines.

(7) FIG. 7 is a diagram illustrating an example of a horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(8) FIG. 8 is a diagram illustrating an example of a horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(9) FIG. 9 is a diagram illustrating an example of a wiring layout within a horizontal plane in the solid-state imaging element in FIG. 1.

(10) FIG. 10 is a diagram illustrating an example of a wiring layout within a horizontal plane in the solid-state imaging element in FIG. 1.

(11) FIG. 11 is a diagram illustrating an example of a wiring layout within a horizontal plane in the solid-state imaging element in FIG. 1.

(12) FIG. 12 is a diagram illustrating an example of a wiring layout within a horizontal plane in the solid-state imaging element in FIG. 1.

(13) FIG. 13 is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(14) FIG. 14 is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(15) FIG. 15 is a diagram illustrating a modification of the horizontal cross-sectional configuration

of the solid-state imaging element in FIG. 1.

(16) FIG. 16 is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(17) FIG. 17 is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(18) FIG. 18 is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(19) FIG. 19 is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element in FIG. 1.

(20) FIG. 20 is a diagram illustrating a modification of the circuitry configuration of the solid-state imaging element configured as in FIG. 1 or according to any of modifications thereof.

(21) FIG. 21 is a diagram illustrating an example of the solid-state imaging element in FIG. 20 including three stacked substrates.

(22) FIG. 22 is a diagram illustrating an example in which the logic circuit is divided to be formed on the substrate on which sensor pixels are disposed and on the substrate on which readout circuits are disposed.

(23) FIG. 23 is a diagram illustrating an example in which the logic circuit is formed on a third substrate.

(24) FIG. 24 is a diagram illustrating part of a cross section of a solid-state imaging element according to a first embodiment of the present disclosure.

(25) FIG. 25 is a schematic diagram illustrating a region around the positions where stacked bodies of the solid-state imaging element according to the first embodiment of the present disclosure are stuck together.

(26) FIG. 26 is a schematic diagram illustrating a configuration of an amplification transistor according to the first embodiment of the present disclosure.

(27) FIG. 27 is a flow diagram illustrating an example of a procedure for manufacturing the solid-state imaging element according to the first embodiment of the present disclosure.

(28) FIG. 28 is a flow diagram illustrating an example of a procedure for manufacturing the solid-state imaging element according to the first embodiment of the present disclosure.

(29) FIG. 29 is a flow diagram illustrating an example of a procedure for manufacturing the solid-state imaging element according to the first embodiment of the present disclosure.

(30) FIG. 30 is a flow diagram illustrating an example of a procedure for manufacturing the solid-state imaging element according to the first embodiment of the present disclosure.

(31) FIG. 31 is a flow diagram illustrating an example of a procedure for manufacturing the solid-state imaging element according to the first embodiment of the present disclosure.

(32) FIG. 32 is a schematic diagram illustrating a solid-state imaging element according to a comparative example of the present disclosure.

(33) FIG. 33 is a schematic diagram illustrating a configuration of an amplification transistor in a solid-state imaging element according to a first modification of the first embodiment of the present disclosure.

(34) FIG. 34 is a diagram illustrating part of a cross section of a solid-state imaging element according to a second modification of the first embodiment of the present disclosure.

(35) FIG. 35 is a diagram illustrating part of a cross section of a solid-state imaging element according to a third modification of the first embodiment of the present disclosure.

(36) FIG. 36 is a schematic diagram illustrating a region around the positions where stacked bodies of a solid-state imaging element according to a second embodiment of the present disclosure are stuck together.

(37) FIG. 37 is a diagram illustrating an example of a schematic configuration of an imaging system including the above-described solid-state imaging element.

(38) FIG. 38 is a diagram illustrating an example of an imaging procedure carried out in the

imaging system in FIG. 37.

(39) FIG. 39 is a diagram illustrating an example of a schematic configuration of an imaging system according to a modification including the above-described solid-state imaging element.

(40) FIG. 40 is a block diagram illustrating an example of a schematic configuration of a vehicle control system.

(41) FIG. 41 is an explanatory diagram illustrating an example of the positions where an imaging unit and a vehicle external information detecting unit are installed.

(42) FIG. 42 is a diagram illustrating an example of a schematic configuration of an endoscopic surgery system.

(43) FIG. 43 is a block diagram illustrating an example of function configurations of a camera head and a CCU.

MODE FOR CARRYING OUT THE INVENTION

(44) Embodiments of the present disclosure will now be described in detail on the basis of the drawings. In the following individual embodiments, identical reference numerals are given to identical parts, and duplicate descriptions are omitted.

(45) [Example Schematic Configuration of Solid-State Imaging Element]

(46) The following describes an example schematic configuration of a solid-state imaging element with reference to FIGS. 1 to 19.

(47) (Example Circuitry Configuration of Solid-State Imaging Element)

(48) FIG. 1 is a diagram illustrating an example schematic configuration of a solid-state imaging element 1 applied to embodiments of the present disclosure. The solid-state imaging element 1 converts received light into an electric signal and outputs the electric signal as a pixel signal. In the illustrated example, the solid-state imaging element 1 is formed to be a complementary metal oxide semiconductor (CMOS) image sensor.

(49) As illustrated in FIG. 1, the solid-state imaging element 1 includes three substrates: a first substrate 10, a second substrate 20, and a third substrate 30. The solid-state imaging element 1 is an imaging device having a three-dimensional structure in which these three substrates are stuck together. The first substrate 10, the second substrate 20, and the third substrate 30 are stacked in the order mentioned.

(50) The first substrate 10 includes a plurality of sensor pixels 12 disposed on a semiconductor substrate 11 to perform photoelectric conversion. The plurality of sensor pixels 12 is provided in a matrix within a pixel region 13 of the first substrate 10. The second substrate 20 includes readout circuits 22 disposed on a semiconductor substrate 21 to output pixel signals based on the electric charge output from the sensor pixels 12, where every single readout circuit 22 is disposed for four sensor pixels 12. The second substrate 20 includes a plurality of pixel drive lines 23 extending along the row direction and a plurality of vertical signal lines 24 extending along the column direction. The third substrate 30 includes a logic circuit 32 disposed on a semiconductor substrate 31 to process pixel signals. The logic circuit 32 includes, for example, a vertical drive circuit 33, a column signal processing circuit 34, a horizontal drive circuit 35, and a system control circuit 36. The logic circuit 32, or more specifically the horizontal drive circuit 35, outputs an output voltage V_{out} for each sensor pixel 12 to the outside. In the logic circuit 32, for example, a low resistance region including a silicide formed by using a self-aligned silicide (SALICIDE) process, such as CoSi₂ or NiSi, may be formed on a surface of the dopant diffusion region in contact with the source electrode and the drain electrode.

(51) The vertical drive circuit 33 selects, for example, a plurality of sensor pixels 12 sequentially row by row. The column signal processing circuit 34 performs, for example, correlated double sampling (CDS) processing on the pixel signals output from the individual sensor pixels 12 in the row selected by the vertical drive circuit 33. For example, the column signal processing circuit 34 extracts signal levels of the pixel signals by performing the CDS processing and holds pixel data corresponding to the amount of light received by each sensor pixel 12. The horizontal drive circuit

35 outputs, for example, the pixel data held in the column signal processing circuit 34 to the outside sequentially. The system control circuit 36 controls, for example, driving of the individual blocks: the vertical drive circuit 33, the column signal processing circuit 34, and the horizontal drive circuit 35 in the logic circuit 32.

(52) FIGS. 2 to 5 each illustrate an example of the sensor pixel 12 and readout circuit 22.

Description is given below regarding a case where a single readout circuit 22 is shared by four sensor pixels 12. The term “shared” here represents that outputs from the four sensor pixels 12 are input to the readout circuit 22 used in common. Note that, however, the unit of sharing is not limited to a specific number of pixels. For example, an output from one sensor pixel 12 may be input to one readout circuit 22. Furthermore, outputs from four sensor pixels 12 may be input to one readout circuit 22, as in the present example.

(53) As illustrated in FIG. 2, each sensor pixel 12 includes components in common with other sensor pixels 12. In FIG. 2, in order to distinguish components among the individual sensor pixels 12, an identification number 1, 2, 3, or 4 is suffixed to the reference symbol of the component of each sensor pixel 12. An identification number is hereinafter suffixed to the reference symbol of a component of each sensor pixel 12 when components need to be distinguished among the individual sensor pixels 12. An identification number to be suffixed to the reference symbol of a component of each sensor pixel 12 is omitted when components do not need to be distinguished among the individual sensor pixels 12.

(54) Each sensor pixel 12 includes, for example, a photodiode PD, a transfer transistor TR that is electrically connected to the photodiode PD, and a floating diffusion FD that temporarily holds the electric charge output from the photodiode PD via the transfer transistor TR. The photodiode PD corresponds to a specific example of a “photoelectric conversion element” of the present disclosure. The photodiode PD performs photoelectric conversion to generate electric charge in accordance with the amount of received light. The cathode of the photodiode PD is electrically connected to the source of the transfer transistor TR, and the anode of the photodiode PD is electrically connected to a reference potential line such as a ground line (GND). The drain of the transfer transistor TR is electrically connected to the floating diffusion FD, and the gate of the transfer transistor TR is electrically connected to the pixel drive line 23 (see FIG. 1). The transfer transistor TR is, for example, a CMOS transistor.

(55) The individual floating diffusions FD of the sensor pixels 12 sharing one readout circuit 22 are electrically connected to one another while being electrically connected to an input terminal of the shared readout circuit 22. The readout circuit 22 includes, for example, a reset transistor RST, a selection transistor SEL, and an amplification transistor AMP. Note that the selection transistor SEL may be omitted if necessary. The source of the reset transistor RST, the source being the input terminal of the readout circuit 22, is electrically connected to the floating diffusion FD, and the drain of the reset transistor RST is electrically connected to a power supply line VDD and to the drain of the amplification transistor AMP. The gate of the reset transistor RST is electrically connected to the pixel drive line 23 (see FIG. 1). The source of the amplification transistor AMP is electrically connected to the drain of the selection transistor SEL, and the gate of the amplification transistor AMP is electrically connected to the source of the reset transistor RST. The source of the selection transistor SEL, the source being the output terminal of the readout circuit 22, is electrically connected to the vertical signal line 24, and the gate of the selection transistor SEL is electrically connected to the pixel drive line 23 (see FIG. 1).

(56) When the transfer transistor TR is turned on, the electric charge in the photodiode PD is transferred to the floating diffusion FD. The reset transistor RST resets the potential of the floating diffusion FD to a predetermined potential. When the reset transistor RST is turned on, the potential of the floating diffusion FD is reset to the potential of the power supply line VDD. The selection transistor SEL controls the timing of outputting a pixel signal from the readout circuit 22. The amplification transistor AMP generates, as a pixel signal, a signal of a voltage corresponding to the

level of the electric charge held in the floating diffusion FD. The amplification transistor AMP, which is included in a source-follower type amplifier, outputs a pixel signal of a voltage corresponding to the level of the electric charge generated in the photodiode PD. When the selection transistor SEL is turned on, the amplification transistor AMP amplifies the potential of the floating diffusion FD and outputs a voltage corresponding to the potential to the column signal processing circuit **34** via the vertical signal line **24**. The reset transistor RST, the amplification transistor AMP, and the selection transistor SEL are, for example, CMOS transistors.

(57) Note that the selection transistor SEL may be disposed between the power supply line VDD and the amplification transistor AMP, as shown in FIG. **3**. In this case, the drain of the reset transistor RST is electrically connected to the power supply line VDD and to the drain of the selection transistor SEL. The source of the selection transistor SEL is electrically connected to the drain of the amplification transistor AMP, and the gate of the selection transistor SEL is electrically connected to the pixel drive line **23** (see FIG. **1**). The source of the amplification transistor AMP, the source being the output terminal of the readout circuit **22**, is electrically connected to the vertical signal line **24**, and the gate of the amplification transistor AMP is electrically connected to the source of the reset transistor RST.

(58) Furthermore, as illustrated in FIGS. **4** and **5**, an FD transfer transistor FDG may be provided between the source of the reset transistor RST and the gate of the amplification transistor AMP. The FD transfer transistor FDG is used for switching the conversion efficiency. In general, low-level pixel signals are generated during imaging in a dark place. On the basis of $Q=CV$, during charge-voltage conversion, a lower voltage V will be generated through conversion in the amplification transistor AMP as the capacitance C of the floating diffusion FD is larger. In contrast, high-level pixel signals are generated during imaging in a bright place, and thus the floating diffusion FD will fail to accept all the electric charge from the photodiode PD unless the FD capacitance C is large. Furthermore, the FD capacitance C needs to be large enough to prevent the voltage V generated through conversion in the amplification transistor AMP from being too high. In view of these matters, when the FD transfer transistor FDG is turned on, the total FD capacitance C is increased due to an additional gate capacitance of the FD transfer transistor FDG. In contrast, when the FD transfer transistor FDG is turned off, the total FD capacitance C is decreased. In this way, by turning on/off the FD transfer transistor FDG, the FD capacitance C can be made variable to switch the conversion efficiency.

(59) FIG. **6** is a diagram illustrating an example of a connection mode between a plurality of readout circuits **22** and a plurality of vertical signal lines **24**. In a case where the plurality of readout circuits **22** is arranged along the column direction in which the vertical signal line **24** extends, each of the plurality of vertical signal lines **24** may be assigned to the readout circuit **22** on a one-to-one basis. For example, as illustrated in FIG. **6**, in a case where four readout circuits **22** are arranged along the direction in which the vertical signal line **24** extends, the four vertical signal lines **24** may be assigned to the readout circuits **22** on a one-to-one basis. Note that each of the vertical signal lines **24** in FIG. **6** is given an identification number 1, 2, 3, or 4 suffixed to the reference symbol for the purpose of distinguishing the vertical signal lines **24** from one another.

(60) (Example Physical Configuration of Solid-State Imaging Element)

(61) FIGS. **7** and **8** are diagrams each illustrating an example of a horizontal cross-sectional configuration of the solid-state imaging element **1**. The upper figure in each of FIGS. **7** and **8** illustrates an example of a horizontal cross-sectional configuration of the first substrate **10** in FIG. **1**. The lower figure in each of FIGS. **7** and **8** illustrates an example of a horizontal cross-sectional configuration of the second substrate **20** in FIG. **1**. FIG. **7** illustrates an example configuration in which two sets of 2×2 four sensor pixels **12** are arranged along a second direction H , while FIG. **8** illustrates an example configuration in which four sets of 2×2 four sensor pixels **12** are arranged along a first direction V and the second direction H . Note that the upper cross-sectional view in each of FIGS. **7** and **8** shows an example of a horizontal cross-sectional configuration of the first

substrate **10** in FIG. **1** overlaid with an example of a surface configuration of the semiconductor substrate **11**. Furthermore, the lower cross-sectional view in each of FIGS. **7** and **8** shows an example of a horizontal cross-sectional configuration of the second substrate **20** in FIG. **1** overlaid with an example of a surface configuration of the semiconductor substrate **21**.

(62) As illustrated in FIGS. **7** and **8**, a plurality of through wirings **54**, a plurality of through wirings **48**, and a plurality of through wirings **47** are arranged zonally in a plane of the first substrate **10** along the first direction V, that is, the vertical direction in FIG. **7**, or the second direction H, that is, the horizontal direction in FIG. **8**. Note that FIGS. **7** and **8** illustrate a case where the plurality of through wirings **54**, the plurality of through wirings **48**, and the plurality of through wirings **47** are arranged in two rows along the first direction V or the second direction H. The first direction V or the second direction H is parallel to, for example, one of the row direction and the column direction, which are the two arrangement directions of the plurality of sensor pixels **12** disposed in a matrix. In the four sensor pixels **12** sharing the readout circuit **22**, the four floating diffusions FD are placed close to one another across a pixel separator **43**, for example. In the four sensor pixels **12** sharing the readout circuit **22**, the gate electrodes TG of the four transfer transistors TR are disposed so as to surround the four floating diffusions FD. For example, the four gate electrodes TG form an annular shape.

(63) An insulating layer **53** present around the above-described plurality of through wirings **54** passing through the semiconductor substrate **21** includes a plurality of blocks extending along the first direction V or the second direction H. The semiconductor substrate **21** includes a plurality of island-like blocks **21A** that extends along the first direction V or the second direction H and is arranged across the above-mentioned insulating layer **53** along the first direction V or the second direction H orthogonal to each other. In each block **21A**, there is provided, for example, a plurality of sets of reset transistors RST, amplification transistors AMP, and selection transistors SEL. One readout circuit **22** shared by four sensor pixels **12** includes, for example, the reset transistor RST, the amplification transistor AMP, and the selection transistor SEL that are disposed in a region facing the four sensor pixels **12**. One readout circuit **22** shared by four sensor pixels **12** includes, for example, the amplification transistor AMP within the block **21A** immediately to the left of the aforementioned insulating layer **53**, and the reset transistor RST and the selection transistor SEL within the block **21A** immediately to the right of the aforementioned insulating layer **53**.

(64) FIGS. **9** to **12** are diagrams each illustrating an example of a wiring layout within a horizontal plane in the solid-state imaging element **1**. FIGS. **9** to **12** each illustrate an example in which one readout circuit **22** shared by four sensor pixels **12** is disposed within a region facing the four sensor pixels **12**. The wirings illustrated in FIGS. **9** to **12** are disposed, for example, in layers different from one another within a wiring layer (not illustrated) provided on the aforementioned pixel transistors. The wiring layer includes, for example, a plurality of pixel drive lines **23**, a plurality of vertical signal lines **24**, and a pad electrode (not illustrated) that is exposed on a surface of the wiring layer and used for electrical connection between the second substrate **20** and the third substrate **30**.

(65) The four through wirings **54** adjacent to one another are electrically connected to, for example, a connection wiring **55**, as illustrated in FIG. **9**. The four through wirings **54** adjacent to one another are further electrically connected to the gate of the amplification transistor AMP included in the block **21A** immediately to the left of the insulating layer **53** and to the gate of the reset transistor RST included in the block **21A** immediately to the right of the insulating layer **53** via, for example, the connection wiring **55** and the connection part **59**.

(66) The power supply line VDD is disposed at, for example, a position facing each of the readout circuits **22** arranged along the second direction H, as illustrated in FIG. **10**. The power supply line VDD is electrically connected to the drain of the amplification transistor AMP and the drain of the reset transistor RST of each of the readout circuits **22** arranged along the second direction H via, for example, the connection part **59**. Two pixel drive lines **23** are each disposed at, for example, a

position facing each of the readout circuits **22** arranged along the second direction H. One of the pixel drive lines **23** is, for example, a wiring RSTG electrically connected to the gate of the reset transistor RST of each of the readout circuits **22** arranged along the second direction H. The other pixel drive line **23** is, for example, a wiring SELG electrically connected to the gate of the selection transistor SEL of each of the readout circuits **22** arranged along the second direction H. In each readout circuit **22**, the source of the amplification transistor AMP and the drain of the selection transistor SEL are electrically connected to each other via, for example, a wiring **25**.

(67) As illustrated in FIG. **11**, two power supply lines VSS are each disposed at, for example, a position facing each of the readout circuits **22** arranged along the second direction H. Each power supply line VSS is electrically connected to a plurality of through wirings **47** at, for example, a position facing each of the sensor pixels **12** arranged along the second direction H. Four pixel drive lines **23** are each disposed at, for example, a position facing each of the readout circuits **22** arranged along the second direction H. Each of the four pixel drive lines **23** is, for example, a wiring TRG electrically connected to the through wiring **48** for a single sensor pixel **12**, which is one of the four sensor pixels **12** corresponding to one of the readout circuits **22** arranged along the second direction H. That is, the four pixel drive lines **23**, which function as control lines, are each electrically connected to the gate electrode TG of the transfer transistor TR for each of the sensor pixels **12** arranged along the second direction H. Note that an identification number 1, 2, 3, or 4 is suffixed to each of the wirings TRG in FIG. **11** for the purpose of distinguishing the wirings TRG from one another.

(68) As illustrated in FIG. **12**, the vertical signal line **24** is disposed at, for example, a position facing each of the readout circuits **22** arranged along the first direction V. The vertical signal line **24**, which functions as an output line, is electrically connected to, for example, the source of the amplification transistor AMP, the source being an output terminal of each of the readout circuits **22** arranged along the first direction V.

First Modification

(69) FIGS. **13** and **14** are diagrams each illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element **1** described above. The upper figure in each of FIGS. **13** and **14** illustrates a modification of the horizontal cross-sectional configuration of the first substrate **10** in FIG. **1**, and the lower figure in FIG. **13** illustrates a modification of the horizontal cross-sectional configuration of the second substrate **20** in FIG. **1**. Note that the upper cross-sectional view in each of FIGS. **13** and **14** shows a modification of the horizontal cross-sectional configuration of the first substrate **10** in FIG. **1** overlaid with a modification of the surface configuration of the semiconductor substrate **11** in FIG. **1**. Furthermore, the lower cross-sectional view in each of FIGS. **13** and **14** shows a modification of the horizontal cross-sectional configuration of the second substrate **20** in FIG. **1** overlaid with a modification of the surface configuration of the semiconductor substrate **21**.

(70) As illustrated in FIGS. **13** and **14**, the plurality of through wirings **54**, the plurality of through wirings **48**, and the plurality of through wirings **47**, which are arranged in a matrix in the form of a plurality of dots, are arranged in a plane of the first substrate **10** zonally along the first direction H, which is the horizontal direction in FIGS. **13** and **14**. Note that FIGS. **13** and **14** illustrate a case where the plurality of through wirings **54**, the plurality of through wirings **48**, and the plurality of through wirings **47** are arranged in two rows along the second direction H. In the four sensor pixels **12** sharing the readout circuit **22**, the four floating diffusions FD are placed close to one another across a pixel separator **43**, for example. In the four sensor pixels **12** sharing the readout circuit **22**, the four transfer electrodes TG1, TG2, TG3, and TG4 are disposed so as to surround the four floating diffusions FD. For example, the four transfer gates TG form an annular shape.

(71) The insulating layer **53** includes a plurality of blocks extending along the second direction H. The semiconductor substrate **21** includes a plurality of island-like blocks **21A** that extends along the second direction H and is arranged across the insulating layer **53** along the first direction V

orthogonal to the second direction H. In each block **21A**, there are provided, for example, the reset transistor RST, the amplification transistor AMP, and the selection transistor SEL. One readout circuit **22** shared by four sensor pixels **12** is disposed, for example, not to entirely face the four sensor pixels **12** but to be shifted along the first direction V.

(72) In FIG. **13**, one readout circuit **22** shared by four sensor pixels **12** includes, on the second substrate **20**, the reset transistor RST, the amplification transistor AMP, and the selection transistor SEL that are disposed in a region shifted along the first direction V from the region facing the four sensor pixels **12**. One readout circuit **22** shared by four sensor pixels **12** includes, for example, the amplification transistor AMP, the reset transistor RST, and the selection transistor SEL that are disposed in a single block **21A**.

(73) In FIG. **14**, one readout circuit **22** shared by four sensor pixels **12** includes, on the second substrate **20**, the reset transistor RST, the amplification transistor AMP, the selection transistor SEL, and the FD transfer transistor FDG that are disposed in a region shifted along the first direction V from the region facing the four sensor pixels **12**. One readout circuit **22** shared by four sensor pixels **12** includes, for example, the amplification transistor AMP, the reset transistor RST, the selection transistor SEL, and the FD transfer transistor FDG that are disposed in a single block **21A**.

(74) In the present modification, one readout circuit **22** shared by four sensor pixels **12** is disposed, for example, not to entirely face the four sensor pixels **12** but is shifted along the first direction V from a position where the readout circuit **22** entirely faces the four sensor pixels **12**. In this case, the wiring **25** (see FIG. **10**) can be shortened, or the wiring **25** can be omitted and a shared doped region may include the source of the amplification transistor AMP and the drain of the selection transistor SEL. As a result, the readout circuit **22** can be made smaller-sized, or other portions in the readout circuit **22** can be made larger-sized.

Second Modification

(75) FIG. **15** is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element **1** described above. FIG. **15** illustrates a modification of the cross-sectional configuration in FIG. **7**.

(76) In the present modification, the semiconductor substrate **21** includes a plurality of island-like blocks **21A** arranged along the first direction V and the second direction H across the insulating layer **53**. In each block **21A**, there is disposed, for example, a single set of the reset transistor RST, the amplification transistor AMP, and the selection transistors SEL. In this case, crosstalk that may occur between readout circuits **22** adjacent to each other can be reduced by the insulating layer **53** to suppress lowering of resolution on a reproduced image or impairing of image quality due to color mixture.

Third Modification

(77) FIG. **16** is a diagram illustrating a modification of the horizontal cross-sectional configuration of the solid-state imaging element **1** described above. FIG. **16** illustrates a modification of the cross-sectional configuration in FIG. **15**.

(78) In the present modification, one readout circuit **22** shared by four sensor pixels **12** is disposed, for example, not to entirely face the four sensor pixels **12** but to be shifted along the first direction V. In addition, in the present modification, the semiconductor substrate **21** includes a plurality of island-like blocks **21A** arranged along the first direction V and the second direction H across the insulating layer **53**, as in the second modification. In each block **21A**, there is disposed, for example, a single set of the reset transistor RST, the amplification transistor AMP, and the selection transistors SEL. In the present modification, a plurality of the through wirings **47** and a plurality of the through wirings **54** are further arranged along the second direction H. Specifically, the plurality of through wirings **47** is disposed between four through wirings **54** sharing a certain readout circuit **22** and four through wirings **54** sharing another readout circuit **22** adjacent to the certain readout circuit **22** along the second direction H. In this case, crosstalk that may occur between readout

circuits **22** adjacent to each other can be reduced by the insulating layer **53** and the through wirings **47** to suppress lowering of resolution on a reproduced image or impairing of image quality due to color mixture.

Fourth Modification

(79) FIG. **17** is a diagram illustrating an example of a horizontal cross-sectional configuration of the solid-state imaging element **1** described above. FIG. **17** illustrates a modification of the cross-sectional configuration in FIG. **7**.

(80) In the present modification, the first substrate **10** has the photodiode PD and the transfer transistor TR disposed for every sensor pixel **12**, and shares the floating diffusion FD for every four sensor pixels **12**. Accordingly, in the present modification, one through wiring **54** is disposed for every four sensor pixels **12**.

(81) Among the plurality of sensor pixels **12** arranged in a matrix, the four sensor pixels **12** corresponding to the region that is obtained by shifting the unit region by a single sensor pixel **12** along the first direction V, the unit region corresponding to the four sensor pixels **12** sharing one floating diffusion FD, are referred to as four sensor pixels **12A** for convenience. Then, in the present modification, the first substrate **10** has the through wiring **47** shared by every four sensor pixels **12A**. Accordingly, in the present modification, one through wiring **47** is disposed for every four sensor pixels **12A**.

(82) In the present modification, the first substrate **10** includes the pixel separator **43** that separates the photodiodes PD and the transfer transistors TR for each sensor pixel **12**. The pixel separator **43** does not completely surround the sensor pixel **12** when viewed from the direction normal to the semiconductor substrate **11**, but includes gaps, which are unformed regions, near the through wiring **54** connected to the floating diffusion FD and near the through wiring **47**. Then, these gaps enable four sensor pixels **12** to share one through wiring **54** and also enable four sensor pixels **12A** to share one through wiring **47**. In the present modification, the second substrate **20** includes the readout circuit **22** for every four sensor pixels **12** sharing the floating diffusion FD.

(83) FIG. **18** is a diagram illustrating an example of a horizontal cross-sectional configuration of the solid-state imaging element **1** according to the present modification. FIG. **18** illustrates a modification of the cross-sectional configuration in FIG. **15**. In the present modification, the first substrate **10** has the photodiode PD and the transfer transistor TR disposed for every sensor pixel **12**, and shares the floating diffusion FD for every four sensor pixels **12**. Moreover, the first substrate **10** includes the pixel separator **43** that separates the photodiodes PD and the transfer transistors TR for each sensor pixel **12**.

(84) FIG. **19** is a diagram illustrating an example of a horizontal cross-sectional configuration of the solid-state imaging element **1** according to the present modification. FIG. **19** illustrates a modification of the cross-sectional configuration in FIG. **16**. In the present modification, the first substrate **10** has the photodiode PD and the transfer transistor TR disposed for every sensor pixel **12**, and shares the floating diffusion FD for every four sensor pixels **12**. Moreover, the first substrate **10** includes the pixel separator **43** that separates the photodiodes PD and the transfer transistors TR for each sensor pixel **12**.

Fifth Modification

(85) FIG. **20** is a diagram illustrating an example of the circuitry configuration of the solid-state imaging element **1** according to a modification. The solid-state imaging element **1** according to the present modification is a CMOS image sensor equipped with a column-parallel ADC.

(86) As illustrated in FIG. **20**, the solid-state imaging element **1** according to the present modification includes the vertical drive circuit **33**, the column signal processing circuit **34**, a reference voltage supply unit **38**, the horizontal drive circuit **35**, a horizontal output line **37**, and the system control circuit **36**, in addition to the pixel region **13** in which the plurality of sensor pixels **12** including photoelectric conversion elements is two-dimensionally arranged in a matrix.

(87) In this system configuration, the system control circuit **36** generates, on the basis of a master

clock MCK, clock signals serving as references for operations of the vertical drive circuit 33, the column signal processing circuit 34, the reference voltage supply unit 38, the horizontal drive circuit 35, and the like, as well as control signals and the like, and supplies the clock signals, the control signals, and the like to the vertical drive circuit 33, the column signal processing circuit 34, the reference voltage supply unit 38, the horizontal drive circuit 35, and the like.

(88) In addition, the vertical drive circuit 33 is formed on the first substrate 10 together with the sensor pixels 12 in the pixel region 13, and is further formed on the second substrate 20 on which the readout circuit 22 is formed. The column signal processing circuit 34, the reference voltage supply unit 38, the horizontal drive circuit 35, the horizontal output line 37, and the system control circuit 36 are formed on the third substrate 30.

(89) Although not illustrated here, a configuration that may be used for the sensor pixel 12 includes the photodiode PD and further includes, for example, the transfer transistor TR that transfers the electric charge obtained by photoelectric conversion in the photodiode PD to the floating diffusion FD. In addition, although not illustrated here, a configuration that may be used for the readout circuit 22 includes, for example, a three-transistor configuration in which the reset transistor RST that controls the potential of the floating diffusion FD, the amplification transistor AMP that outputs a signal in accordance with the potential of the floating diffusion FD, and the selection transistor SEL for selecting a pixel are disposed.

(90) In the pixel region 13, the sensor pixels 12 are two-dimensionally arranged, and to the m rows by n columns arrangement, the pixel drive line 23 is wired for each row and the vertical signal line 24 is wired for each column. One end of each of the plurality of pixel drive lines 23 is connected to each output end corresponding to each row in the vertical drive circuit 33. The vertical drive circuit 33, which includes a shift register and others, controls row addresses and row scanning in the pixel region 13 through the plurality of pixel drive lines 23.

(91) The column signal processing circuit 34 includes, for example, analog-to-digital conversion circuits (ADCs) 34-1 to 34-m each provided for each pixel column in the pixel region 13, that is, for each vertical signal line 24, to convert an analog signal output from each sensor pixel 12 in each column in the pixel region 13 into a digital signal and outputs the digital signal.

(92) The reference voltage supply unit 38 includes, for example, a digital-to-analog conversion circuit (DAC) 38A as a method for generating a reference voltage Vref having a so-called ramp (RAMP) waveform in which the level varies in an inclined manner over time. Note that the method for generating a reference voltage Vref having a ramp waveform is not limited to the DAC 38A.

(93) Under the control of control signals CS1 given by the system control circuit 36, the DAC 38A generates a reference voltage Vref having a ramp waveform on the basis of the clock CK given by the system control circuit 36, and supplies the reference voltage Vref to the ADCs 34-1 to 34-m of the column processing unit 15.

(94) Note that each of the ADCs 34-1 to 34-m is configured to be able to selectively perform the AD conversion operation in an operation mode supporting both the normal frame rate mode in which all the information in the sensor pixel 12 is read by using a progressive scan method and the high-speed frame mode in which the frame rate is N-fold (double, for example) as compared with the normal frame mode by setting the exposure time of the sensor pixel 12 to 1/N. The operation mode is switched under the control of control signals CS2 and CS3 given by the system control circuit 36. Furthermore, to the system control circuit 36, instruction information for switching the operation mode between the normal frame rate mode and the high-speed frame rate mode is given by an external system controller (not illustrated).

(95) The ADCs 34-1 to 34-m each have the same configuration, and thus the description below takes the ADC 34-m as an example. The ADC 34-m includes a comparator 34A, an up/down counter (U/D CNT) 34B, which is an example of counting means, a transfer switch 34C, and a memory device 34D.

(96) The comparator 34A compares the signal voltage Vx of the vertical signal line 24 with the

reference voltage V_{ref} , where the signal voltage V_x corresponds to a signal output from each sensor pixel **12** in the n -th column in the pixel region **13** and the reference voltage V_{ref} has a ramp waveform and is supplied from the reference voltage supply unit **38**. For example, when the reference voltage V_{ref} is higher than the signal voltage V_x , the output V_{co} is the “H” level, and when the reference voltage V_{ref} is equal to or lower than the signal voltage V_x , the output V_{co} is the “L” level.

(97) The up/down counter **34B**, which is an asynchronous counter, is given the clock CK by the system control circuit **36** at the same time as the DAC **18A** under the control of control signals CS2 given by the system control circuit **36**, and counts down (DOWN) or counts up (UP) in synchronization with the clock CK to measure the comparison time period from the start to the end of a comparison operation performed by the comparator **34A**.

(98) Specifically, in the normal frame rate mode, for the operation of reading out a signal from one sensor pixel **12**, the up/down counter **34B** counts down during the first readout operation to measure the comparison time in the first readout operation, and counts up during the second readout operation to measure the comparison time in the second readout operation.

(99) On the other hand, in the high-speed frame rate mode, the count result with respect to the sensor pixel **12** in a certain row is held, and subsequently, with respect to the sensor pixel **12** in the next row, the up/down counter **34B** counts down from the previous count result during the first readout operation to measure the comparison time in the first readout operation, and counts up during the second readout operation to measure the comparison time in the second readout operation.

(100) In the normal frame rate mode, upon completion of the count operation by the up/down counter **34B** with respect to the sensor pixel **12** in a certain row, the transfer switch **34C** is turned on (closed) and transfers the result of counting by the up/down counter **34B** to the memory device **34D**, under the control of control signals CS3 given by the system control circuit **36**.

(101) On the other hand, in the high-speed frame rate of, for example, $N=2$, when the count operation by the up/down counter **34B** is completed with respect to the sensor pixel **12** in a certain row, the transfer switch **34C** remains off (open). Subsequently, when the count operation by the up/down counter **34B** is completed with respect to the sensor pixel **12** in the next row, the transfer switch **34C** is turned on and transfers the result of counting by the up/down counter **34B** for the vertical two pixels to the memory device **34D**.

(102) In this way, an analog signal supplied from each sensor pixel **12** in the pixel region **13** column by column through the vertical signal lines **24** is converted into an N -bit digital signal through operations by the comparator **34A** and the up/down counter **34B** in each of the ADCs **34-1** to **34- m** , and the digital signal is stored in the memory device **34D**.

(103) The horizontal drive circuit **35**, which includes a shift register and others, controls column addresses and column scanning on ADCs **34-1** to **34- m** in the column signal processing circuit **34**. Under the control of the horizontal drive circuit **35**, N -bit digital signals resulting from AD conversion in the individual ADCs **34-1** to **34- m** are sequentially read out to the horizontal output line **37**, and then output as imaging data through the horizontal output line **37**.

(104) In addition to the above-described components, there may be provided, for example, a circuit (not illustrated because the circuit is not directly related to the present disclosure) that performs various types of signal processing on the imaging data that is output through the horizontal output line **37**.

(105) In the solid-state imaging element **1** equipped with the column-parallel ADC according to the present modification configured as above, the result of counting by the up/down counter **34B** can be selectively transferred to the memory device **34D** through the transfer switch **34C**, and therefore, the count operation by the up/down counter **34B** and the operation of reading out the result of counting by the up/down counter **34B** to the horizontal output line **37** can be independently controlled.

(106) FIG. 21 illustrates an example of the solid-state imaging element 1 in FIG. 20 including stacked substrates: the first substrate 10, the second substrate 20, and the third substrate 30.

(107) In the present modification, the pixel region 13 including the plurality of sensor pixels 12 is formed in a central portion on the first substrate 10, and the vertical drive circuit 33 is formed around the pixel region 13.

(108) In addition, on the second substrate 20, the readout circuit region 15 including the plurality of readout circuits 22 is formed in a central portion and the vertical drive circuit 33 is formed around the readout circuit region 15.

(109) Furthermore, on the third substrate 30, the column signal processing circuit 34, the horizontal drive circuit 35, the system control circuit 36, the horizontal output line 37, and the reference voltage supply unit 38 are formed.

(110) Owing to the configuration above, as in the aforementioned configuration in FIG. 1 and modifications thereof, there is no increase in chip size or no inhibition on making the area per pixel much smaller, which may be caused by a structure in which substrates are electrically connected. As a result, the solid-state imaging element 1 in a three-layer structure can be provided on a chip equal to conventional chips in size without inhibiting the area per pixel from being much smaller. Note that the vertical drive circuit 33 may be formed only on the first substrate 10 or only on the second substrate 20.

Sixth Modification

(111) FIG. 22 illustrates a modification of the cross-sectional configuration of the solid-state imaging element 1 according to the present modification. In the aforementioned configuration in FIG. 1 and modifications thereof, the solid-state imaging element 1 includes three stacked substrates: the first substrate 10, the second substrate 20, and the third substrate 30. However, in the above configuration in FIG. 1 and modifications thereof, the solid-state imaging element 1 may include two stacked substrates: the first substrate 10 and the second substrate 20.

(112) In this case, the logic circuit 32 is, for example, formed separately on the first substrate 10 and the second substrate 20 as illustrated in FIG. 22. Note here that, in a circuit 32A of the logic circuit 32 disposed on the first substrate 10 side, there is provided a transistor having a gate structure in which a high dielectric constant film including a material being able to withstand a high-temperature process (high-k, for example) and a metal gate electrode are stacked. On the other hand, in a circuit 32B disposed on the second substrate 20 side, a low resistance region including a silicide formed by using a self-aligned silicide (SALICIDE) process, such as CoSi₂ or NiSi, is formed on a surface of the dopant diffusion region in contact with the source electrode and the drain electrode. The low resistance region containing a silicide thus includes a compound of the material of the semiconductor substrate and a metal.

(113) Therefore, a high-temperature process such as thermal oxidation can be used to form the sensor pixel 12. Furthermore, in a case where the low resistance region 26 containing a silicide is disposed on a surface of the dopant diffusion region in contact with the source electrode and the drain electrode in the circuit 32B, which is part of the logic circuit 32 and disposed on the second substrate 20 side, the contact resistance can be reduced. As a result, calculations in the logic circuit 32 can be performed at higher speed.

(114) FIG. 23 is a diagram illustrating a modification of the cross-sectional configuration of the solid-state imaging element 1 configured as in FIG. 1 or according to any of modifications thereof as described above. In the logic circuit 32 on the third substrate 30 configured as in FIG. 1 or according to any of modifications thereof as described above, a low resistance region 37 containing a silicide formed by using a self-aligned silicide (SALICIDE) process, such as CoSi₂ or NiSi, may be formed on a surface of the dopant diffusion region in contact with the source electrode and the drain electrode. Therefore, a high-temperature process such as thermal oxidation can be used to form the sensor pixel 12. Furthermore, in a case where the low resistance region 37 containing a silicide is disposed on a surface of the dopant diffusion region in contact with the source electrode

and the drain electrode in the logic circuit **32**, the contact resistance can be reduced. As a result, calculations in the logic circuit **32** can be performed at higher speed.

First Embodiment

(115) A solid-state imaging element of a first embodiment will now be described with reference to FIGS. **24** to **33**.

(116) (Example Overall Configuration of Solid-State Imaging Element)

(117) FIG. **24** is a diagram illustrating part of a cross section of a solid-state imaging element **100** according to the first embodiment of the present disclosure. As illustrated in FIG. **24**, the solid-state imaging element **100** has a structure in which a stacked body **200**, a stacked body **300**, and a stacked body **400** are stuck together. A plane **230** shown in FIG. **24** is the plane on which the stacked body **200** and the stacked body **300** are stuck together. Furthermore, a plane **340** shown in FIG. **24** is the plane on which the stacked body **300** and the stacked body **400** are stuck together. These stacked bodies **200** to **400** are electrically connected to one another.

(118) A color filter **211** is disposed below the stacked bodies **200** to **400**, that is, on the bottom of the stacked body **200**. An on-chip lens **212** is disposed under the color filter **211**. The on-chip lens **212** condenses the incident light. The condensed light is guided through the color filter **211** to a photoelectric conversion element **203** included in the stacked body **200**.

(119) The stacked body **200** has a structure in which multiple films including a transistor and others are stacked on a substrate **201**. The substrate **201** is a semiconductor substrate such as an N-type silicon substrate. On the substrate **200**, a P-type semiconductor region **202** (P well), for example, is formed. An N-type semiconductor region is formed in the semiconductor region **202**, whereby the photoelectric conversion element **203**, such as a photodiode having a PN junction, is formed. The photoelectric conversion element **203** converts, through photoelectric conversion, the received light into an electric signal corresponding to the amount of the received light.

(120) A hole accumulation diode (HAD) **204**, which is a P.sup.+ type semiconductor region, is formed above the photoelectric conversion element **203**. The HAD **204** functions as a hole accumulation layer to suppress a dark current generated on the surface of the photoelectric conversion element **203**, which is an N-type photodiode.

(121) On the substrate **201**, an N-type transfer transistor **220** is disposed. The transfer transistor **220** includes a floating diffusion (FD) **221**, which is an N-type source region. The transfer transistor **220** transfers an electric signal, which is output from the photoelectric conversion element **203**, to a pixel transistor. The FD **221** temporarily holds an electric signal output from the photoelectric conversion element **203**.

(122) Both the transfer transistor **220** including the FD **221** and the HAD **204** are covered with an insulating film **250**.

(123) The stacked body **300** has a structure in which multiple films including a transistor and others are stacked on a substrate **301**. The substrate **301** is a semiconductor substrate such as a P-type silicon substrate. The stacked body **300** is turned upside down and stuck on the insulating film **250** in the stacked body **200**.

(124) On the substrate **301**, that is, on the side of the substrate **301** facing the substrate **201**, pixel transistors including an amplification transistor **310** of N type, a reset transistor **320** of N type, and a selection transistor of N type (not illustrated) are disposed. The pixel transistors perform a process of reading out an electric signal corresponding to the amount of light received by the photoelectric conversion element **203**.

(125) A wiring **313d** is connected to a gate electrode **313** of the amplification transistor **310**. The wiring **313d** is connected to a source region **321** of the reset transistor **320**. Furthermore, the wiring **313d** is connected to the FD **221** of the transfer transistor **320** via a contact **221c**.

(126) The pixel transistors including the amplification transistor **310** and the reset transistor **320** are covered with an insulating film **350**. Accordingly, the insulating film **250** and the insulating film **350** are joined together on the plane **230** on which the stacked body **200** and the stacked body **300**

are stuck together.

(127) On the lower surface of the substrate **301**, that is, on the side opposite to the side on which the pixel transistors are disposed, wirings **D1** to **D4** are formed in four layers. The wiring **D1** is formed in the lowermost first layer. The wiring **D4** is formed in the uppermost fourth layer. Note that the number of wiring layers is not limited to four but can be changed to any number in accordance with design conditions or the like.

(128) The wirings **D1** to **D4** are covered with an insulating film **360**.

(129) The stacked body **400** has a structure in which multiple films including a transistor and others are stacked on a substrate **401**. The substrate **401** is a semiconductor substrate such as a silicon substrate. The stacked body **400** is turned upside down and joined onto the wiring **D4** in the stacked body **300**. In the example in FIG. **24**, a junction point **402** between the wiring **D4** and a wiring in the stacked body **400** is superimposed on a pixel region in which pixels are disposed.

(130) A plurality of logic transistors **Tr** disposed on the substrate **401**, that is, on the side of the substrate **401** opposite to the side facing the substrate **301** is connected to the wirings in the stacked body **400**. The wirings in the stacked body **400** and the logic transistors **Tr** are covered with an insulating film **450**. The wirings in the stacked body **400** and the logic transistors **Tr** constitute a logic circuit. The logic circuit is a peripheral circuit disposed in the solid-state imaging element **100** to process, for example, the electric signals generated in the photoelectric conversion element **203**.

(131) (Example Detailed Configuration of Solid-State Imaging Element)

(132) The following describes an example detailed configuration of the solid-state imaging element **100** of the first embodiment with reference to FIG. **25**. FIG. **25** is a schematic diagram illustrating a region around the position where the stacked bodies **200** and **300** of the solid-state imaging element **100** according to the first embodiment of the present disclosure are stuck together. FIG. **25(a)** is a top view of the stacked body **300** on the side where the pixel transistors are formed, and FIGS. **25(b)** to **25(d)** are cross-sectional views showing regions around the position where the stacked bodies **200** and **300** are stuck together. FIG. **25(b)** is a cross-sectional view taken along line A-A' in FIG. **25(a)**, FIG. **25(c)** is a cross-sectional view taken along line B-B' in FIG. **25(a)**, and FIG. **25(d)** is a cross-sectional view taken along line C-C' in FIG. **25(a)**. Note that the insulating film **350** and the contact **221c** are omitted in FIG. **25(a)**. In addition, the position of the contact **223c** is shifted in FIG. **25(b)**.

(133) As illustrated in FIG. **25(c)**, the solid-state imaging element **100** includes the substrate **201**, which is a first semiconductor substrate including the FD **221** that temporarily holds an electric signal output from the photoelectric conversion element **203**. The HAD **204** is disposed above the photoelectric conversion element **203**. A contact **204c** connected to the upper layer wiring is connected to the HAD **204**. The contact **204c** is grounded via the upper layer wiring to fix the substrate potential of the substrate **201** to 0 V. The FD **221** is the source region of the transfer transistor **220**. The transfer transistor **220** includes a gate insulating film **224** disposed on the substrate **201** and a gate electrode **223** disposed on the gate insulating film **224**. The contact **223c** connected to the upper layer wiring is connected to the gate electrode **223**. The contact **223c** is connected to the peripheral circuit including the logic transistor **Tr** via the upper layer wiring and wirings in the stacked body **400**. The transfer transistor **220** transfers an electric signal, which is output from the photoelectric conversion element **203**, to the amplification transistor **310**.

(134) The solid-state imaging element **100** includes the substrate **301**, which is a second semiconductor substrate facing the substrate **201**. The substrate **301** includes the amplification transistor **310**, which is a first transistor, on the side facing the substrate **201**. The amplification transistor **310** includes a channel **315** extending along the thickness direction of the substrate **301** and a gate electrode **313**, which is a multi-gate extending along the thickness direction of the substrate **301** and sandwiching the channel **315**. The channel **315**, which includes part of the substrate **301**, serves as a current path between a source region **311** and a drain region **312**, which are described later, when a voltage is applied to the gate electrode **313**. A gate insulating film **314** is

interposed between the channel **315** and the gate electrode **313**. The amplification transistor **310** is configured as, for example, a tri-gate transistor in which the gate electrode **313** is connected on three sides to the channel **315** via the gate insulating film **314**. The amplification transistor **310** amplifies the electric signal transferred from the photoelectric conversion element **203** by the transfer transistor **220** and outputs the amplified electric signal.

(135) As illustrated in FIG. 25(d), the substrate **301** includes the reset transistor **320**, which is a second transistor including a source region **322**, on the side facing the substrate **201**. The reset transistor **320** includes a channel **325** extending along the thickness direction of the substrate **301** and a gate electrode **323**, which is a multi-gate extending along the thickness direction of the substrate **301** and sandwiching the channel **325**. The channel **325**, which includes part of the substrate **301**, serves as a current path between a source region **321** and a drain region **322**, which are described later, when a voltage is applied to the gate electrode **313**. A gate insulating film **324** is interposed between the channel **325** and the gate electrode **323**. The reset transistor **320** is configured as, for example, a tri-gate transistor in which the gate electrode **323** is connected on three sides to the channel **325** via the gate insulating film **324**. The reset transistor **320** resets (initializes) the potential of the gate electrode **313** of the amplification transistor **310** to the power supply potential. The reset transistor **320** also serves as a transistor that resets the potential of the FD **221**.

(136) The gate electrode **323** of the reset transistor **320** is connected to the wirings D1 to D4, which are signal lines transmitting electrical signals from the surface side of the substrate **301** opposite to the surface facing the substrate **201**. Specifically, the gate electrode **323** is connected to the wirings D1 to D4 via a contact **323c**. The wirings D1 to D4 are connected to the peripheral circuit including the logic transistor Tr via wirings in the stacked body **400** to send and receive electric signals.

(137) As illustrated in FIG. 25(b), the substrate **301** includes a selection transistor **330** on the side facing the substrate **201**. The selection transistor **330** includes a channel **335** extending along the thickness direction of the substrate **301** and a gate electrode **333**, which is a multi-gate extending along the thickness direction of the substrate **301** and sandwiching the channel **335**. The channel **335**, which includes part of the substrate **301**, serves as a current path between a source region **331** and a drain region **332**, which are described later, when a voltage is applied to the gate electrode **313**. A gate insulating film **334** is interposed between the channel **335** and the gate electrode **333**. The selection transistor **330** is configured as, for example, a tri-gate transistor in which the gate electrode **333** is connected on three sides to the channel **335** via the gate insulating film **334**. The selection transistor **330** selects whether or not to transmit an electric signal amplified in the amplification transistor **310** to the wirings D1 to D4 in the upper layer so that the electric signal is processed.

(138) As illustrated in FIG. 25(a), the gate electrode **313** of the amplification transistor **310** and the gate electrode **333** of the selection transistor **330** are disposed in parallel. The gate electrode **333** of the selection transistor **330** and the gate electrode **323** of the reset transistor **320** are disposed to be orthogonal to each other.

(139) As illustrated in FIG. 25(c), the gate electrode **313** of the amplification transistor **310** is connected to the FD **221**. Specifically, the solid-state imaging element **100** includes a contact **221c** that connects the gate electrode **313** and the FD **221** between their surfaces facing each other. That is, in the example in FIG. 25, the contact **221c** containing polysilicon or the like connects the surface of the gate electrode **313** extending along the thickness direction of the substrate **301** toward the substrate **201**, the surface being the closest to the substrate **201**, and the outermost surface of the surface layer of the substrate **201** where the FD **221** is disposed. In other words, the contact **221c** connects the gate electrode **313** and the FD **221** in a shortest distance.

(140) The gate electrode **313** of the amplification transistor **310** is connected to the source region **321** of the reset transistor **320**. Specifically, the gate electrode **313** of the amplification transistor **310** extends toward the reset transistor **320** to form the wiring **313d**. The gate electrode **313** of the

amplification transistor **310** and the source region **321** of the reset transistor **320** are connected by the wiring **313d**.

(141) As illustrated in FIG. 25(b), the substrate **301** includes source regions **311** and **331** that extend from one surface side to the other surface side of the substrate **301**, and also includes drain regions **312**, **322**, and **333** that extend from one surface side to the other surface side of the substrate **301**. The source region **311** and the drain region **312** have N-type conductivity with a doping concentration of, for example, $1 \times 10^{18} \text{ cm}^{-3}$ or higher, and are included in the amplification transistor **310**. The source region **331** and the drain region **332** have N-type conductivity with a doping concentration of, for example, $1 \times 10^{18} \text{ cm}^{-3}$ or higher, and are included in the selection transistor **330**. The drain region **332** of the selection transistor **330** is connected to the source region **311** of the amplification transistor **310**. The drain region **322** has N-type conductivity with a doping concentration of, for example, $1 \times 10^{18} \text{ cm}^{-3}$ or higher, and is included in the reset transistor **320**.

(142) Note here that the source region **321** of the reset transistor **320** also has N-type conductivity. However, the source region **321** of the reset transistor **320** is formed only in the surface layer on the surface side of the substrate **301** facing the substrate **201**, without reaching the opposite surface of the substrate **301**. The region extending from the FD **221** to reach the source region **321** of the reset transistor **320** via the contact **221c**, the gate electrode **313** of the amplification transistor **310**, and the wiring **313d** is an FD region functioning as a floating diffusion. The source region **321** is made smaller than others in order to avoid an increase in the FD capacitance.

(143) The source regions **311** and **331** are connected to the wirings **D1** to **D4**, which are signal lines transmitting electrical signals from the surface side of the substrate **301** opposite to the surface facing the substrate **201**. Specifically, the source region **311** is connected to the wirings **D1** to **D4** via a contact **311c**. The source region **331** is connected to the wirings **D1** to **D4** via a contact **331c**. The wirings **D1** to **D4** are connected to the peripheral circuit including the logic transistor **Tr** via wirings in the stacked body **400** to send and receive electric signals.

(144) The drain regions **312**, **322**, and **333** are connected to the power supply potential from the surface side of the substrate **301** opposite to the surface facing the substrate **201**. Specifically, the drain region **312** is connected to the wirings **D1** to **D4** via a contact **312c**. The drain region **322** is connected to the wirings **D1** to **D4** via a contact **322c**. The drain region **332** is connected to the wirings **D1** to **D4** via a contact **332c**. The wirings **D1** to **D4** are connected to the power supply potential.

(145) (Example Detailed Configuration of Gate Electrode)

(146) As described above, the pixel transistors disposed on the substrate **301** are configured to be, for example, tri-gate transistors. The structure of a tri-gate transistor is described here in more detail with reference to FIG. 26, taking the amplification transistor **310** as an example. The reset transistor **320** and the selection transistor **330** are configured in a similar manner to the amplification transistor **310** as described below.

(147) FIG. 26 is a schematic diagram illustrating a configuration of the amplification transistor **310** according to the first embodiment of the present disclosure. FIG. 26(a) is an exploded perspective view of the amplification transistor **310**, and FIG. 26(b) is a perspective view of the amplification transistor **310**.

(148) As illustrated in FIG. 26, the source region **311**, the drain region **312**, and the channel **314** sandwiched therebetween are formed into a plate upright along the stacking direction **SD** of the stacked body **300**.

(149) Part of the source region **311**, all of the channel **314**, and part of the drain region **312** are covered with the gate insulating film **314**. The gate insulating film **314** includes, for example, a high-k material such as Al_2O_3 , TiO_2 , Ta_2O_5 , HfO_2 , HfSiON , HfSiO_4 , ZrO_2 , ZrSiO_4 , La_2O_3 , or Y_2O_3 .

(150) The gate insulating film **314** is covered with the gate electrode **313**. The gate electrode **313**

includes, for example, polysilicon. The amplification transistor **313** may be a metal gate transistor in which the gate electrode **313** includes a metal-based material such as TaCx, W, WNx, or TiN.

(151) In the amplification transistor **310** configured as a tri-gate transistor, the gate width is the sum of the width of the plate-shaped channel (thickness of the plate) and its height $\times 2$.

(152) As described above, the amplification transistor **310** includes the source region **311** of N type, the drain region **312** of N type, and the channel **315** of P type sandwiched between these regions. In addition, the insulating film **360** is disposed directly below the NPN-structured body of the amplification transistor **310**. That is, the amplification transistor **310** has a fully depleted silicon-on-insulator (FD-SOI) structure.

(153) (Example Process of Manufacturing Solid-State Imaging Element)

(154) The following describes an example process of manufacturing the solid-state imaging element **100** of the first embodiment with reference to FIGS. **27** to **31**. FIGS. **27** to **31** are flow diagrams illustrating an example of a procedure for manufacturing the solid-state imaging element **100** according to the first embodiment of the present disclosure. Note that the figures on the left in FIGS. **27** to **31** are cross-sectional views of the in-process solid-state imaging element **100** taken along line A-A' in FIG. **25(a)**. The figures in the center in FIGS. **27** to **31** are cross-sectional views of the in-process solid-state imaging element **100** taken along line B-B' in FIG. **25(a)**. The figures on the right in FIGS. **27** to **31** are cross-sectional views of the in-process solid-state imaging element **100** taken along line C-C' in FIG. **25(a)**.

(155) As illustrated in FIGS. **27(a1)**, **27(b1)**, and **27(c1)**, the substrate **301**, which is a P-type silicon substrate or the like, is subjected to element isolation, and then the channels **315**, **325** are formed by making a trench TR. Although not illustrated, the channel **335** is also formed at the same time.

(156) As illustrated in FIGS. **27(a2)**, **27(b2)**, and **27(c2)**, the gate insulating films **314**, **324**, and **334** are formed so as to cover the channels **315**, **325**, and **335**. Furthermore, the gate electrodes **313**, **323**, and **333** are formed so as to cover the gate insulating films **314**, **324**, and **334**.

(157) Then, the source regions of N type **311**, **321**, and **331** and the drain regions of N type **312**, **322**, and **332** are formed in the substrate **301** on both sides of the gate electrodes **313**, **323**, and **333** so as to have a doping concentration of 1×10^{18} cm.⁻³ or higher. The source regions **311** and **331** and the drain regions **312**, **322**, and **332** are formed to the depth of the trench TR. The source region **321** is formed to be shallower than the other source regions **311** and **331**.

(158) As illustrated in FIGS. **27(a3)**, **27(b3)**, and **27(c3)**, while the insulating film **350** is stacked over the individual components on the substrate **301**, the wiring **313d** connecting the gate electrode **313** and the source region **321** is formed. The insulating film **350** is stacked until the components including the wiring **313d** are entirely covered.

(159) As illustrated in FIGS. **28(a1)**, **28(b1)**, and **28(c1)**, in the substrate **201**, which is an N-type silicon substrate or the like, the semiconductor region **202** of P type is formed, the photoelectric conversion element **203**, which is an N-type photodiode or the like, is formed, and the HAD **204**, which is a semiconductor region of P.⁺ type, is formed.

(160) Furthermore, the gate insulating film **224** is formed on the substrate **201**, and the gate electrode **223** is formed on the gate insulating film **224**. Then, as an N-type source region, the FD **221** is formed in the substrate **201** near the gate electrode **223**.

(161) After that, the insulating film **250** is formed on the substrate **201** so as to cover the individual components. The substrate **301** described above is turned upside down and placed so that the surface on which pixel transistors are formed faces the substrate **201**, which has the components formed thereon.

(162) As illustrated in FIGS. **28(a2)**, **28(b2)**, and **28(c2)**, the substrate **201** and the substrate **301** are stuck together. In this process, the insulating film **250** formed on the substrate **201** and the insulating film **350** formed on the substrate **301** are joined together.

(163) As a result, the transfer transistor **220** on the substrate **201** and the pixel transistors on the substrate **301** face each other. In addition, the wiring **313d** extended from the gate electrode **313** is

disposed directly above the FD 221 on the substrate 201.

(164) As illustrated in FIGS. 29(a1), 29(b1), and 29(c1), the substrate 301 is made thinner by grinding the surface of the substrate 301 opposite to the side on which the pixel transistors are formed. The substrate 301 is made thinner until, for example, the bulk substrate 301 disappears, the end of each of the channels 315, 325, and 335 on the side opposite to the side covered with the gate electrodes 313, 323, and 333 and the like is exposed, both ends of each of the U-shaped gate insulating films 314, 324, and 334 are exposed, and both ends of each of the U-shaped gate electrodes 313, 323, and 333 are exposed. Note that, however, the bulk substrate 301 may be left. As illustrated in FIGS. 29(a1), 29(b1), and 29(c1), in a case where the bulk substrate 301 is eliminated, each pixel transistor has an FD-SOI structure.

(165) Note that, although the substrate 301 that has been ground partly remains separated around each of the pixel transistors, such a substrate 301 is not illustrated in the following drawings.

(166) From the surface of the substrate 301 opposite to the side on which the pixel transistors are formed as illustrated in FIGS. 29(a2), 29(b2), and 29(c2), ion implantation or the like is carried out at positions corresponding to the source regions 311 and 331 and the drain regions 312, 322, and 332 so as to have a doping concentration of 1×10^{18} cm.^{sup.}−3 or higher. As a result, the source regions 311 and 331 and the drain regions 312, 322, and 332 that extend from one surface side of the substrate 301 to reach the other surface side are obtained.

(167) As illustrated in FIGS. 30(a1), 30(b1), and 30(c1), the insulating film 360 is formed on the substrate 301 to cover the individual components. Then, a through hole TH passing through the insulating films 360, 350, and 250 and the wiring 313d to reach the FD 221 in the substrate 201 is formed.

(168) As illustrated in FIGS. 30(a2), 30(b2), and 30(c2), the through hole TH is filled with a conductive material such as polysilicon up to the height of the wiring 313d to form the contact 221c connecting the wiring 313d and the FD 221.

(169) As illustrated in FIGS. 31(a1), 31(b1), and 31(c1), the insulating films 350 and 360 above the wiring 313d are backfilled with an insulating material such as SiO₂.

(170) As illustrated in FIGS. 31(a2), 31(b2), and 31(c2), the contact 223c is formed on the gate electrode 223 and connected to the upper layer wiring. The contact 204c is formed on the HAD 204 and connected to the upper layer wiring.

(171) Furthermore, the contact 323c is formed on the gate electrode 323 and connected to the wirings D1 to D4. Although not illustrated, the contact 333c is also formed on the gate electrode 333 and connected to the wirings D1 to D4.

(172) Furthermore, the contacts 311c and 331c are formed on the source regions 311 and 331 and connected to the wirings D1 to D4. The contacts 312c, 322c, and 332c are formed on the drain regions 312, 322, and 332 and connected to the wirings D1 to D4.

(173) After that, the stacked body 400 in which the peripheral circuit including the logic transistor Tr, the wirings, and others are formed is stuck onto the stacked body 300. In this process, the insulating film 450 in the stacked body 400 and the insulating film 360 in the stacked body 300 are joined together. In addition, the wiring in the stacked body 400 and the wiring D4 in the stacked body 300 are connected. As a result, the wirings D1 to D4 are appropriately connected to the peripheral circuit, the ground line, the power supply potential, and others in the stacked body 400.

(174) The process of manufacturing the solid-state imaging element 100 of the first embodiment is now finished.

Comparative Example

(175) With reference to FIG. 32, a comparison is made below between a configuration of a comparative example and the configuration of the first embodiment. FIG. 32 is a schematic diagram illustrating a solid-state imaging element according to the comparative example of the present disclosure.

(176) In the solid-state imaging element according to Patent Document 1, the semiconductor

substrate on which a pixel region is formed and the semiconductor substrate on which a logic circuit is formed are joined together. That is, photoelectric conversion elements and pixel transistors are formed on the same semiconductor substrate. However, such a configuration fails to secure a sufficient space for pixel transistors to be disposed. For example, in a case where an amplification transistor among the pixel transistors is small in size, it is difficult to increase the transconductance g_m or to reduce noise to a sufficient extent.

(177) Then, a conceivable solution may be, for example, sub-dividing a substrate into a substrate on which the photoelectric conversion elements are formed and a substrate on which the pixel transistors are formed, and joining these substrates together. Such a configuration is shown in FIG. 32 as a comparative example.

(178) As illustrated in FIG. 32, the solid-state imaging element of the comparative example includes a transfer transistor **220'** that includes an FD **221'** on a substrate **201'** that includes a photoelectric conversion element **203'** and an HAD **204'**. A substrate **301'** is disposed above the substrate **201'**. On the upper surface of the substrate **301'**, that is, on the surface opposite to the substrate **201'**, an amplification transistor **310'**, a reset transistor **320'**, and a selection transistor **330'** are disposed. These pixel transistors are planar transistors. In addition, the gate electrode of the amplification transistor **310'**, the source region of the reset transistor **320'**, and the FD **221'** are connected via the contact **221c'** and the wiring **D1'**.

(179) However, such a configuration makes it necessary to extend the contact **221c'** to the level of the wiring **D1'**, resulting in a greater total wiring length. Furthermore, the configuration for connecting the gate electrode of the amplification transistor **310'**, the source region of the reset transistor **320'**, and the FD **221'** is complicated. Therefore, the capacity of wirings involved in the FD **221'** is increased, and the capacitance of the entire FD region is also increased. As a result, the photoelectric conversion efficiency of the photoelectric conversion element **203'** is decreased.

(180) In the solid-state imaging element **100** of the first embodiment, each pixel transistor is configured as a tri-gate transistor and is disposed to face the substrate **201**. As a result, the gate electrode **313** of the amplification transistor **310** and the FD **221** can be brought closer to each other. Furthermore, the source region **321** of the reset transistor **320** and the FD **221** can be brought closer to each other. Therefore, the length of the entire wirings involved in the FD **221**, that is, the lengths of the contact **221c** and the wiring **313d** can be reduced to improve the photoelectric conversion efficiency of the photoelectric conversion element **203**.

(181) In the solid-state imaging element **100** of the first embodiment, each pixel transistor is configured as a tri-gate transistor. Therefore, while the amplification transistor **310** faces the substrate **201**, both ends of the U-shaped gate electrode **323** of the reset transistor **320** are allowed to face the wirings **D1** to **D4**. As a result, the gate electrode **323** can be connected to the wirings **D1** to **D4** from the side of the substrate **301** facing the wirings **D1** to **D4**.

(182) In the solid-state imaging element **100** of the first embodiment, each pixel transistor is configured as a tri-gate transistor. As a result, the gate width of the pixel transistor can be expanded in the direction perpendicular to the surface of the substrate **301** without increasing the area occupied in the substrate **301**, thereby further reducing noise and improving the transconductance g_m .

(183) In the solid-state imaging element **100** of the first embodiment, each pixel transistor has an FD-SOI structure. As a result, pixel transistors can be miniaturized while the parasitic capacitance can be reduced to obtain high-speed pixel transistors.

(184) The solid-state imaging element **100** of the first embodiment includes the source regions **311** and **331** and the drain regions **312**, **322**, and **332** distributed all across the thickness direction of the substrate **301**. Therefore, while the pixel transistors face the substrate **201**, the source regions **311** and **331** and the drain regions **312**, **322**, and **332** can be connected to the wirings **D1** to **D4** from the side of the substrate **301** facing the wirings **D1** to **D4**. Therefore, the configuration for connecting the source regions **311** and **331** and the drain regions **312**, **322**, and **332** to the wirings **D1** to **D4** is

not complicated. Furthermore, since each pixel transistor is a tri-gate transistor, the gate electrodes **313**, **323**, and **333** have greater control over the channels **315**, **325**, and **335**. Therefore, even when the source regions **311** and **331** and the drain regions **312**, **322**, and **332** having a high doping concentration are distributed from the lower surface to the upper surface of the substrate **301**, a short circuit between the source regions **311**, **321**, and **331** and the drain regions **312**, **322**, and **332** can be suppressed.

(185) The above-described configuration allows the solid-state imaging element **100** of the first embodiment to fully take advantage of disposing the photoelectric conversion element **203** and the pixel transistor on the separate substrates **201** and **301**. That is, as compared with the case where the photoelectric conversion element and the pixel transistors are disposed on the same substrate, the areas of both the photoelectric conversion element **203** and the pixel transistor can be made larger. Furthermore, the number of pixels per unit area can be increased.

(186) Moreover, in the solid-state imaging element **100** of the first embodiment, the substrate **201** and the substrate **301** are connected via the contact **221c**. Furthermore, the substrate **301** and the substrate **401** are connected by the wiring **D3** in the substrate **301** and the wiring in the substrate **401**. These configurations reduce an area required for connection between substrates, as compared with the case where substrates are connected by a through silicon via (TSV) disposed in a peripheral region of the substrates. As a result, the chip size for the solid-state imaging element **100** can be reduced. Alternatively, the pixel region can be made larger with the same chip size.

(187) In addition, in the solid-state imaging element **100** of the first embodiment, the junction point **402** of the contact **221c** and the wiring **D3** in the substrate **301** with the wiring in the substrate **401** is disposed within the pixel region. As a result, the chip size can be further reduced or the pixel region can be made larger.

First Modification

(188) The following describes a solid-state imaging element according to a first modification of the first embodiment with reference to FIG. **33**. FIG. **33** is a schematic diagram illustrating a configuration of an amplification transistor in the solid-state imaging element according to the first modification of the first embodiment of the present disclosure. The amplification transistor of the first modification is a multi-gate transistor of a type different from that of the first embodiment. A reset transistor of the first modification and a selection transistor of the first modification are configured in a similar manner to the amplification transistor described below.

(189) As illustrated in FIG. **33(a)**, an amplification transistor **310a** of the first modification is configured as a double-gate transistor in which a gate electrode **313a** is connected to a channel on two sides via a gate insulating film **314a**. That is, the amplification transistor **310a** includes a source region **311a** of N type, an N-type drain region (not illustrated), and a P-type channel sandwiched therebetween (not illustrated).

(190) Both side surfaces of part of the source region **311a**, both side surfaces of the entire channel, and both side surfaces of part of the drain region are covered with the gate insulating film **314a**. The gate insulating film **314a** includes a high-k material or the like as in the first embodiment. The lower end of part of the source region **311a**, the lower end of the entire channel, and the lower end of part of the drain region in the figure are covered with an insulating film **316in**.

(191) The gate insulating film **314a** and the insulating film **316in** are covered with the gate electrode **313a**. From the gate electrode **313a**, a wiring **313da** is extended to be connected to the source region of the reset transistor or the like. As in the first embodiment, the gate electrode **313a** and the wiring **313da** include polysilicon or a metal-based material or the like.

(192) In the amplification transistor **310a** configured as a double-gate transistor, the gate width is twice the height of the plate-shaped channel.

(193) The amplification transistor **310a** of the first modification can also be configured as a transistor having an FD-SOI structure in which the insulating film **360** is disposed directly below the NPN-structured body.

(194) As illustrated in FIG. 33(b), an amplification transistor **310b** of the first modification is configured as an all-around transistor having a gate-all-around (GAA) structure in which a gate electrode **313b** is connected on four sides to a channel via a gate insulating film **314b**. That is, the amplification transistor **310b** includes a source region **311b** of N type, an N-type drain region (not illustrated), and a P-type channel sandwiched therebetween (not illustrated).

(195) The source region **311b**, the channel, and the drain region are in a shape of a plate upright with respect to the insulating film **360**. The source region **311b** includes a wing portion **311w** that is bent in a V shape and is in contact with the insulating film **360**. The drain region includes a wing portion (not illustrated) that is bent in a V shape and is in contact with the insulating film **360**.

(196) The perimeter of part of the source region **311b**, the perimeter of the entire channel, and the perimeter of part of the drain region are covered with the gate insulating film **314b**. The gate insulating film **314b** includes a high-k material or the like as in the first embodiment.

(197) The gate insulating film **314b** is covered with the gate electrode **313b**. From the gate electrode **313b**, a wiring **313db** is extended to be connected to the source region of the reset transistor or the like. As in the first embodiment, the gate electrode **313b** and the wiring **313db** include polysilicon or a metal-based material or the like.

(198) In the amplification transistor **310b** configured as an all-around transistor, the length of the perimeter of the plate-shaped channel is the gate width.

(199) The amplification transistor **310b** of the first modification can also be configured as a transistor having an FD-SOI structure in which the insulating film **360** is disposed directly below the NPN-structured body.

(200) As examples of the pixel transistor, the foregoing has shown a tri-gate transistor in the first embodiment and a double-gate transistor and an all-around transistor in the first modification; however, configurations of the pixel transistor are not limited thereto. Any type of the pixel transistor may be selected from various types of multi-gate transistors.

(201) As a result, it is made possible to form a pixel transistor that can take a contact to both the substrate on which the photoelectric conversion element is formed and the wiring on the upper layer above the pixel transistor.

(202) As a result, the pixel transistor is also allowed to have greater control over the channel. Therefore, it is made possible to form a source region and a drain region that can take a contact to both the substrate on which the photoelectric conversion element is formed and the wiring on the upper layer above the pixel transistor, while suppressing a short circuit between the source region and the drain region.

Second Modification

(203) The following describes a solid-state imaging element **110** according to a second modification of the first embodiment with reference to FIG. 34. FIG. 34 is a diagram illustrating part of a cross section of the solid-state imaging element **110** according to the second modification of the first embodiment of the present disclosure.

(204) As illustrated in FIG. 34, in the solid-state imaging element **110** of the second modification, a gate electrode **223x** of the transfer transistor **220** is extended to reach the photodiode **203**. That is, the transfer transistor **220** may include the gate electrode **223x** formed to be a vertical transfer gate.

Third Modification

(205) The following describes a solid-state imaging element **120** according to a third modification of the first embodiment with reference to FIG. 35. FIG. 35 is a diagram illustrating part of a cross section of the solid-state imaging element **120** according to the third modification of the first embodiment of the present disclosure.

(206) As illustrated in FIG. 35, in the solid-state imaging element **120** of the third modification, the stacked body **300** and the stacked body **400** are electrically connected in a region opposed to a peripheral region **14** of the stacked body **200**. The peripheral region **14**, which corresponds to the frame region of the stacked body **200**, is provided on a peripheral edge of the pixel region **13**. The

stacked body **300** includes a plurality of pad electrodes **58** in a region opposed to the peripheral region **14**, and the stacked body **400** includes a plurality of pad electrodes **64** in a region opposed to the peripheral region **14**. The stacked body **300** and the stacked body **400** are electrically connected to each other by joining the pad electrodes **58** and **64** provided in a region opposed to the peripheral region **14**.

(207) In this way, the stacked body **300** and the stacked body **400** are connected by joining the pad electrodes **58** and **64**. Therefore, as compared with the case where two stacked bodies are connected by, for example, a TSV disposed in a peripheral region of the stacked bodies, the chip size can be made smaller or the pixel region can be made larger.

Second Embodiment

(208) A solid-state imaging element of a second embodiment will now be described with reference to FIG. **36**. FIG. **36** is a schematic diagram illustrating a region around the position where stacked bodies of the solid-state imaging element according to the second embodiment of the present disclosure are stuck together. The solid-state imaging element of the second embodiment is different from the solid-state imaging element of the above-described first embodiment in that a selection transistor **530** is disposed on a substrate **501**, which is different from the substrate on which an amplification transistor **310e** and others are disposed.

(209) Note that FIG. **36(a)** is a cross-sectional view taken along line A-A' in FIG. **25(a)**, which is the cited drawing, FIG. **36(b)** is a cross-sectional view taken along line B-B' in FIG. **25(a)**, and FIG. **36(c)** is a cross-sectional view taken along line C-C' in FIG. **25(a)**.

(210) As illustrated in FIG. **36**, the solid-state imaging element of the second embodiment includes the stacked body **200**, a stacked body **300e** stuck on the stacked body **200**, and a stacked body **500** stuck on the stacked body **300e**.

(211) The substrate **301e** on the stacked body **300e** does not have any selection transistor. That is, the substrate **301e**, which is a P-type silicon substrate or the like, includes the amplification transistor **310e** and the reset transistor **320**.

(212) The amplification transistor **310e** is, for example, a tri-gate transistor that includes a source region **311e** of N type, a drain region **312e** of N type, a channel **315e** of P type, a gate insulating film **314e**, and a gate electrode **313e**. However, the amplification transistor **310e** may be another multi-gate transistor such as a double-gate transistor or an all-around gate transistor. The amplification transistor **310e** is formed to be larger than, for example, the amplification transistor **310** of the first embodiment by the size of the selection transistor absent on the substrate **301e**.

(213) The solid-state imaging element of the second embodiment includes the substrate **501**, which is a third semiconductor substrate disposed on the side opposite to the substrate **201**, which is the first semiconductor substrate, so as to face the substrate **301e**, which is the second semiconductor substrate. That is, the stacked body **300e** and the stacked body **500** including the substrate **501** are joined together on a plane **355** between the insulating film **360** covering the substrate **301e** and the insulating film **550** covering the substrate **501**.

(214) The substrate **501**, which is a P-type silicon substrate or the like, includes the selection transistor **530** that selects whether or not to transmit the electric signal amplified in the amplification transistor **510e** to the wirings **D1** to **D4** serving as signal lines. The selection transistor **530** is disposed on the surface opposite to the surface facing the substrate **301e**. The selection transistor **530**, which includes, for example, a source region **531**, a channel **535**, and a drain region **532** that are disposed on the surface layer of the substrate **501**, is formed to be a planar transistor that includes a gate insulating film **534** on the substrate **501** and a gate electrode **533** on the gate insulating film **534**.

(215) The drain region **532** of the selection transistor **530** is connected to the source region **311e** of the amplification transistor **510** via a contact **532c**, the wiring **D2**, and the contact **311c**. The source region **531** of the selection transistor **530** is connected to the upper layer wiring via a contact **531c**.

(216) In the solid-state imaging element of the second embodiment, the selection transistor **530** is

disposed on the substrate **501**, which is a substrate separate from the substrate **301e**. As a result, the amplification transistor **310e** on the substrate **301e** can be made much larger, whereby noise can be further reduced and transconductance g_m can be further improved.

(217) Note that, the selection transistor **530** is a planar transistor in the configuration of the second embodiment, but this is not restrictive. The selection transistor may be configured as a multi-gate transistor such as a tri-gate transistor, as in the first embodiment and others. In this case, the source region and the drain region of the selection transistor can be distributed all across the thickness direction of the substrate on which the selection transistor is formed. Therefore, the drain region and the source region **311e** of the amplification transistor **310e** can be connected between the facing surfaces. Furthermore, the source region and the upper layer wiring can be connected between the facing surfaces. The selection transistor may be orientated upward or downward for the connection.

Third Embodiment

(218) FIG. **37** is a diagram illustrating an example of a schematic configuration of an imaging system **2** that includes any of the solid-state imaging elements of the first and second embodiments and modifications thereof. Accordingly, any of the solid-state imaging elements of the first and second embodiments and modifications thereof can be disposed in the imaging system **2**. The following description takes the imaging system **2** that includes the solid-state imaging element **100** of the first embodiment as an example.

(219) The imaging system **2** serving as a video recording apparatus is, for example, an imaging apparatus such as a digital still camera or a video camera, or an electronic device such as a portable terminal apparatus like a smart phone or a tablet-type terminal. The imaging system **2** includes, for example, the solid-state imaging element **100** of the first embodiment, a DSP circuit **141**, a frame memory **142**, a display unit **143**, a storage unit **144**, an operation unit **145**, and a power supply unit **146**. In the imaging system **2**, the solid-state imaging element **100**, the DSP circuit **141**, the frame memory **142**, the display unit **143**, the storage unit **144**, the operation unit **145**, and the power supply unit **146** are connected to one another via a bus line **147**.

(220) The solid-state imaging element **100** outputs image data in accordance with incident light. The DSP circuit **141** is a signal processing circuit that processes image data, which is a signal output from the solid-state imaging element **100**. The frame memory **142** temporarily holds, on a frame basis, the image data subjected to the processing in the DSP circuit **141**. The display unit **143**, which includes, for example, a panel-type display device such as a liquid crystal panel or an organic electro luminescence (EL) panel, displays a moving image or a still image captured by the solid-state imaging element **100**. The storage unit **144** records the image data containing a moving image or a still image captured by the solid-state imaging element **100** on a recording medium such as a semiconductor memory or a hard disk. The operation unit **145** issues operation commands for various functions of the imaging system **2** in accordance with a user operation. The power supply unit **146** appropriately provides various power sources serving as operating power sources for the solid-state imaging element **100**, the DSP circuit **141**, the frame memory **142**, the display unit **143**, the storage unit **144**, and the operation unit **145** to these supply targets.

(221) The following describes an imaging procedure carried out in the imaging system **2**.

(222) FIG. **38** shows an example flowchart for imaging operations in the imaging system **2**. The imaging system **2** accepts the start of imaging caused by an operation or the like of the operation unit **145** by the user (step **S101**). Then, the operation unit **145** sends an imaging command to the imaging element **1** (step **S102**). Upon receipt of the imaging command, a system control circuit (see, for example, the system control circuit **36** in FIG. **1**) of the imaging element **100** carries out imaging based on a predetermined imaging method (step **S103**).

(223) The solid-state imaging element **100** outputs the image data obtained by imaging to the DSP circuit **141**. The image data here refers to the data representing pixel signals for all the pixels as generated on the basis of the electric charge temporarily held in the floating diffusion FD. The DSP

circuit **141** performs predetermined signal processing such as noise reduction on the basis of the image data input from the solid-state imaging element **100** (step **S104**). The DSP circuit **141** causes the frame memory **142** to hold the image data subjected to the predetermined signal processing, and the frame memory **142** causes the storage unit **144** to store the image data (step **S105**). In this way, imaging is carried out in the imaging system **2**.

(224) Since the imaging system **2** includes the smaller-sized or higher-definition solid-state imaging element **100**, the smaller-sized or higher-definition imaging system **2** can be provided.

Modifications

(225) FIG. **39** is a diagram illustrating an example of a schematic configuration of an imaging system **201** according to a modification including any of the solid-state imaging elements of the first and second embodiments and modifications thereof. That is, the imaging system **201** is a modification of the imaging system **2** described above. The following description takes the imaging system **201** that includes the solid-state imaging element **100** of the first embodiment as an example.

(226) As shown in FIG. **39**, the imaging apparatus **201** includes an optical system **202**, a shutter device **203**, the solid-state imaging element **100**, a control circuit **205**, a signal processing circuit **206**, a monitor **207**, and a memory **208**, and is capable of capturing still images and moving images.

(227) The optical system **202**, which includes one or more lenses, guides the light (incident light) from a subject to the solid-state imaging element **100** to form an image on a light receiving surface of the solid-state imaging element **100**.

(228) The shutter device **203**, which is disposed between the optical system **202** and the solid-state imaging element **100**, controls the time period when the light is incident on the solid-state imaging element **100** and the time period when the light is shielded, under the control of the control circuit **205**.

(229) The solid-state imaging element **100** accumulates signal charge for a certain period of time in accordance with the light to be turned into an image formed on the light receiving surface via the optical system **202** and the shutter device **203**. The signal charge accumulated in the solid-state imaging element **100** is transferred in accordance with a drive signal (timing signal) supplied from the control circuit **205**.

(230) The control circuit **205** drives the solid-state imaging element **100** and the shutter device **203** by outputting drive signals for controlling the transfer operation of the solid-state imaging element **100** and the shuttering operation of the shutter device **203**.

(231) The signal processing circuit **206** performs various types of signal processing on the signal charge output from the solid-state imaging element **100**. The image (image data) obtained through the signal processing performed by the signal processing circuit **206** is supplied to and displayed on the monitor **207** or supplied to and stored (recorded) in the memory **208**.

(232) The imaging system **201** configured as above is also capable of imaging on every pixel with low noise by applying the solid-state imaging element **100** thereto.

Application Example 1

(233) The technology according to the present disclosure can be applied to various products. For example, the technology according to the present disclosure may be implemented in a device provided in any type of mobile object selected from an automobile, an electric vehicle, a hybrid electric vehicle, a motorcycle, a bicycle, a personal mobility, an airplane, a drone, a ship, a robot, and the like.

(234) FIG. **40** is a block diagram illustrating a schematic configuration example of a vehicle control system, which is an example of a mobile body control system to which the technology according to the present disclosure can be applied.

(235) The vehicle control system **12000** includes a plurality of electronic control units connected via a communication network **12001**. In the example illustrated in FIG. **40**, the vehicle control

system **12000** includes a drive system control unit **12010**, a body system control unit **12020**, a vehicle external information detecting unit **12030**, a vehicle internal information detecting unit **12040**, and an integrated control unit **12050**. In addition, as a functional configuration of the integrated control unit **12050**, a microcomputer **12051**, an audio image output unit **12052**, and an in-vehicle network interface (I/F) **12053** are illustrated.

(236) The drive system control unit **12010** controls operations of devices related to the drive system of the vehicle in accordance with various programs. For example, the drive system control unit **12010** functions as a control device for a driving force generation device for generating a driving force for an internal combustion engine, a driving motor, or the like, of the vehicle, a driving force transmission mechanism for transmitting a driving force to a wheel, a steering mechanism for adjusting a steering angle of the vehicle, a braking device for generating a braking force of the vehicle, and the like.

(237) The body system control unit **12020** controls operations of various devices provided in the vehicle body in accordance with various programs. For example, the body system control unit **12020** functions as a control device for a keyless entry system, a smart key system, a power window device, or for various lamps such as a head lamp, a back lamp, a brake lamp, a blinker, a fog lamp, or the like. In this case, radio waves transmitted from a mobile device that substitutes for a key or signals of various switches may be input to the body system control unit **12020**. The body system control unit **12020** accepts the radio waves or signals that have been input, and controls the door lock device, the power window device, the lamps, and the like of the vehicle.

(238) The vehicle external information detecting unit **12030** detects information regarding the outside of the vehicle in which the vehicle control system **12000** is installed. For example, an imaging unit **12031** is connected to the vehicle external information detecting unit **12030**. The vehicle external information detecting unit **12030** causes the imaging unit **12031** to capture an image of the outside of the vehicle and receives the captured image. On the basis of the received image, the vehicle external information detecting unit **12030** may perform a process of detecting an object, such as a person, a vehicle, an obstacle, a sign, or a character on a road surface, or a process of detecting a distance.

(239) The imaging unit **12031** is an optical sensor that receives light and outputs an electric signal corresponding to the amount of the light received. The imaging unit **12031** can output electric signals representing an image, and also can output electric signals representing information regarding distance measurement. Furthermore, the light to be received by the imaging unit **12031** may be visible light or invisible light such as infrared light.

(240) The vehicle internal information detecting unit **12040** detects information regarding the inside of the vehicle. For example, a driver state detecting unit **12041** that detects the state of the driver is connected to the vehicle internal information detecting unit **12040**. The driver state detecting unit **12041** may include, for example, a camera that captures images of the driver. On the basis of the detection information that is input from the driver state detecting unit **12041**, the vehicle internal information detecting unit **12040** may calculate the degree of fatigue or the degree of concentration of the driver or may determine whether or not the driver is dozing.

(241) The microcomputer **12051** is capable of calculating a control target value for the driving force generation device, the steering mechanism, or the braking device on the basis of the information regarding the inside or outside of the vehicle as acquired by the vehicle external information detecting unit **12030** or the vehicle internal information detecting unit **12040**, and of outputting a control command to the drive system control unit **12010**. For example, the microcomputer **12051** is capable of performing cooperative control for fulfilling functions of advanced driver assistance systems (ADAS) including collision avoidance or shock mitigation of the vehicle, following travel based on an inter-vehicle distance, vehicle-speed maintained traveling, vehicle collision warning, vehicle lane departure warning, and the like.

(242) Furthermore, the microcomputer **12051** is capable of performing cooperative control for

automated driving and the like in which the vehicle autonomously travels without relying on driver's operations, by controlling the driving force generation device, the steering mechanism, the braking device, or the like on the basis of the information regarding surroundings of the vehicle as acquired by the vehicle external information detecting unit **12030** or the vehicle internal information detecting unit **12040**.

(243) Furthermore, the microcomputer **12051** is capable of outputting a control command to the body system control unit **12020** on the basis of the information regarding the outside of the vehicle as acquired by the vehicle external information detecting unit **12030**. For example, the microcomputer **12051** is capable of performing cooperative control to prevent dazzling by, for example, controlling the headlamps in accordance with the position of a preceding vehicle or oncoming vehicle detected by the vehicle external information detecting unit **12030** to switch from the high beam to the low beam.

(244) The audio image output unit **12052** transmits an output signal, which is at least one of a sound or an image, to an output device that can visually or auditorily give notification of information to an occupant of the vehicle or to the outside of the vehicle. In the example in FIG. **40**, an audio speaker **12061**, a display unit **12062**, and an instrument panel **12063** are illustrated as the output device. The display unit **12062** may include, for example, at least one of an on-board display or a head-up display.

(245) FIG. **41** is diagram showing example positions at which the imaging unit **12031** is disposed.

(246) In FIG. **41**, the vehicle **12100** includes imaging units **12101**, **12102**, **12103**, **12104**, and **12105** as the imaging unit **12031**.

(247) The imaging units **12101** to **12105** are disposed in, for example, a front nose, a side mirror, a rear bumper, a back door, an upper portion of a windshield in a vehicle interior, and the like of the vehicle **12100**. The imaging unit **12101** disposed in the front nose and the imaging unit **12105** disposed in an upper portion of the windshield in the vehicle interior mainly acquire images of a region in front of the vehicle **12100**. The imaging units **12102** and **12103** disposed in the side mirrors mainly acquire images of a region on either side of the vehicle **12100**. The imaging unit **12104** disposed in the rear bumper or the back door mainly acquires images of a region on the back of the vehicle **12100**. Images of a front region acquired by the imaging units **12101** and **12105** are mainly used for detecting a preceding vehicle, a pedestrian, an obstacle, a traffic signal, a traffic sign, a lane, or the like.

(248) Note that FIG. **41** shows an example of imaging ranges of the imaging units **12101** to **12104**. An imaging range **12111** indicates an imaging range of the imaging unit **12101** disposed in the front nose, imaging ranges **12112** and **12113** indicate imaging ranges of the imaging units **12102** and **12103**, respectively, disposed in the side mirrors, and an imaging range **12114** indicates an imaging range of the imaging unit **12104** disposed in the rear bumper or the back door. For example, pieces of image data captured by the imaging units **12101** to **12104** are superimposed so that a bird's-eye image of the vehicle **12100** seen from above can be obtained.

(249) At least one of the imaging units **12101** to **12104** may have a function of acquiring distance information. For example, at least one of the imaging units **12101** to **12104** may be a stereo camera including a plurality of imaging elements or may be an imaging element having pixels to detect a phase difference.

(250) For example, by obtaining a distance to each of solid objects in the imaging ranges **12111** to **12114** and a temporal change in the distance, that is, relative speed to the vehicle **12100**, on the basis of the distance information acquired from the imaging units **12101** to **12104**, the microcomputer **12051** can extract a solid object that is on the traveling route of the vehicle **12100**, positioned closest to the vehicle **12100**, and traveling at a predetermined speed, for example, equal to or higher than 0 km/h, in a direction substantially the same as the direction in which the vehicle **12100** is traveling, as a preceding vehicle. Moreover, the microcomputer **12051** can set a distance between vehicles to be secured beforehand in front of the preceding vehicle and perform automatic

brake control including following stop control, automatic acceleration control including following start control, and the like. In this way, cooperative control can be performed for automated driving and the like by which the vehicle autonomously travels without relying on driver's operations. (251) For example, on the basis of the distance information acquired from the imaging units **12101** to **12104**, the microcomputer **12051** can classify solid object data regarding solid objects into two-wheeled vehicle, regular vehicle, large vehicle, pedestrian, and other solid objects such as a utility pole, to extract data, and can use the extracted data to automatically avoid an obstacle. For example, the microcomputer **12051** identifies an obstacle around the vehicle **12100** either as an obstacle that can be visually recognized by the driver of the vehicle **12100** or as an obstacle that is difficult for the driver to visually recognize. Then, the microcomputer **12051** determines a collision risk indicating the degree of risk of collision with each obstacle and, when the collision risk is equal to or higher than a predetermined value and the vehicle can possibly collide with the obstacle, the microcomputer **12051** can assist driving for collision avoidance by outputting a warning to the driver via the audio speaker **12061** or the display unit **12062** or by performing forced deceleration or avoidance steering via the drive system control unit **12010**.

(252) At least one of the imaging units **12101** to **12104** may be an infrared camera that detects infrared light. For example, the microcomputer **12051** can recognize a pedestrian by determining whether or not a pedestrian is present in images captured by the imaging units **12101** to **12104**. Recognition of a pedestrian is performed by, for example, a procedure of extracting feature points in images captured by the imaging units **12101** to **12104** as an infrared camera and a procedure of performing a pattern matching process on a series of feature points indicating the contour of an object to determine whether or not the object is a pedestrian. When the microcomputer **12051** determines that a pedestrian is present in images captured by the imaging units **12101** to **12104** and recognizes the pedestrian, the audio image output unit **12052** causes the display unit **12062** to display a superimposed rectangular outline to emphasize the recognized pedestrian. Furthermore, the audio image output unit **12052** may cause the display unit **12062** to display an icon or the like indicating the pedestrian at a desired position.

(253) The foregoing has described an example of the mobile body control system to which the technology according to the present disclosure may be applied. The technology according to the present disclosure may be applied to the imaging unit **12031** among the above-described components. Specifically, the solid-state imaging element according to any of the above-described first and second embodiments and modifications thereof can be applied to the imaging unit **12031**. Applying the technology according to the present disclosure to the imaging unit **12031** provides high-definition and low-noise captured images. Therefore, the mobile body control system can perform control with high precision using the captured images.

Application Example 2

(254) FIG. **42** is a diagram illustrating a schematic configuration example of an endoscopic surgery system to which the technology according to the present disclosure can be applied.

(255) FIG. **42** shows that an operator **11131** such as a surgeon is performing surgery on a patient **11132** on an examination table **11133** using an endoscopic surgery system **11000**. As illustrated in the figure, the endoscopic surgery system **11000** includes an endoscope **11100**, other surgical tools **11110** such as a pneumoperitoneum tube **11111** and an energy treatment tool **11112**, a support arm device **11120** that supports the endoscope **11100**, and a cart **11200** containing various devices for endoscopic surgery.

(256) The endoscope **11100** includes a lens barrel **11101** whose tip is to be inserted into a body cavity of the patient **11132** by a predetermined length and a camera head **11102** connected to the base end of the lens barrel **11101**. In the illustrated example, the endoscope **11100** is formed to be a so-called rigid scope having the lens barrel **11101** being rigid, but the endoscope **11100** may be formed to be a so-called flexible scope having a flexible lens barrel.

(257) The tip of the lens barrel **11101** is provided with an opening into which an objective lens is

fitted. A light source device **11203** is connected to the endoscope **11100**. The light generated by the light source device **11203** is guided to the tip of the lens barrel **11101** by a light guide extending inside the lens barrel **11101**, and is shed on the observation target in the body cavity of the patient **11132** via the objective lens. Note that the endoscope **11100** may be a forward-viewing endoscope, an oblique-viewing endoscope, or a side-viewing endoscope.

(258) The camera head **11102** contains an optical system and any of the solid-state imaging elements according to the above-described first and second embodiments and modifications thereof. The light reflected from the observation target, that is, the observation light, is condensed by the optical system onto the solid-state imaging element. The solid-state imaging element performs photoelectric conversion on the observation light to generate an electric signal corresponding to the observation light, that is, an image signal corresponding to the observation image. The image signal, which is treated as RAW data, is sent to a camera control unit (CCU) **11201**.

(259) The CCU **11201**, which includes a central processing unit (CPU), a graphics processing unit (GPU), and the like, comprehensively controls operations of the endoscope **11100** and a display device **11202**. Moreover, the CCU **11201** receives an image signal from the camera head **11102** and performs various types of image processing on the image signal for displaying an image based on the image signal, such as development processing (demosaijing).

(260) Under the control of the CCU **11201**, the display device **11202** displays an image based on the image signal that has undergone image processing by the CCU **11201**.

(261) The light source device **11203**, which includes a light source such as a light emitting diode (LED), supplies irradiation light to the endoscope **11100** when a surgical site is imaged.

(262) An input device **11204** is an input interface for the endoscopic surgery system **11000**. The user can input various types of information and instructions to the endoscopic surgery system **11000** via the input device **11204**. For example, the user inputs an instruction or the like to change imaging conditions such as the type of irradiation light, magnification, and focal length provided by the endoscope **11100**.

(263) The treatment tool control device **11205** controls driving of the energy treatment tool **11112** for cauterizing or incising a tissue, blocking a blood vessel, or the like. For the purpose of securing a field of view of the endoscope **11100** and securing a working space for the operator **11131**, the pneumoperitoneum device **11206** sends gas into the body cavity of the patient **11132** via the pneumoperitoneum tube **11111** to inflate the body cavity. The recorder **11207** is a device capable of recording various types of information regarding the surgery. The printer **11208** is a device capable of printing various types of information regarding the surgery in various forms such as text, images, and graphs.

(264) Note that the light source device **11203** that supplies irradiation light to the endoscope **11100** when the surgical site is imaged may include a white light source that includes, for example, an LED, a laser light source, or a combination thereof. In a case where the white light source includes a combination of RGB laser light sources, the output intensity of each wavelength in each color and the output timing can be controlled with high accuracy. Therefore, the white balance of a captured image can be adjusted in the light source device **11203**. Furthermore, in this case, images corresponding to R, G, and B can be captured by time division by irradiating the observation target with laser light beams from their corresponding RGB laser light sources by time division and controlling driving of the solid-state imaging element in the camera head **11102** in synchronization with the irradiation timing. By using this method, a color image can be obtained without providing the solid-state imaging element with a color filter.

(265) In addition, the light source device **11203** may be controlled to be driven so as to change the intensity of the output light at predetermined time intervals. By acquiring images by time division and combining the images through controlled driving of the solid-state imaging element in the camera head **11102** in synchronization with the timing when the light intensity is changed, a high

dynamic-range image can be generated without so-called blown-out highlights or blocked-up shadows.

(266) Furthermore, the light source device **11203** may be configured to be capable of supplying light of a predetermined wavelength band suitable for special light observation. Special light observation includes, for example, carrying out so-called narrow band imaging, which uses dependency of a body tissue on wavelengths for light absorption. In narrow band imaging, the light having a narrower band than white light, which is the incident light for usual observations, is shed on a body tissue to image a certain tissue, such as blood vessels in a mucous membrane surface layer, with high contrast. Special light observation may also include fluorescence observation in which an image is obtained by fluorescence generated by emitting excitation light. Fluorescence observation may include, for example, auto-fluorescence observation in which a body tissue is irradiated with excitation light to observe fluorescence produced in the body tissue, or obtaining a fluorescence image by locally injecting a reagent such as indocyanine green (ICG) into a body tissue while irradiating the body tissue with the excitation light corresponding to the fluorescence wavelength of the reagent. The light source device **11203** may be configured to be capable of supplying at least one of the narrow band light or the excitation light suitable for such special light observation.

(267) FIG. **43** is a block diagram illustrating an example of functional configurations of the camera head **11102** and the CCU **11201** shown in FIG. **42**.

(268) The camera head **11102** includes a lens unit **11401**, an imaging unit **11402**, a drive unit **11403**, a communication unit **11404**, and a camera head control unit **11405**. The CCU **11201** includes a communication unit **11411**, an image processing unit **11412**, and a control unit **11413**. The camera head **11102** and the CCU **11201** are communicatively connected to each other by a transmission cable **11400**.

(269) The lens unit **11401** is an optical system provided in a portion connected to the lens barrel **11101**. The observation light taken from the tip of the lens barrel **11101** is guided to the camera head **11102** and enters the lens unit **11401**. The lens unit **11401** includes a plurality of combined lenses including a zoom lens and a focus lens.

(270) The imaging unit **11402** includes a solid-state imaging element. The solid-state imaging element included in the imaging unit **11402** may be one element of so-called single-plate type or a plurality of elements of so-called multi-plate type. In a case where the imaging unit **11402** includes multi-plate type solid-state imaging elements, the individual solid-state imaging elements may generate image signals respectively corresponding to R, G, and B and the image signals may be combined so that a color image can be obtained. Alternatively, the imaging unit **11402** may include a pair of solid-state imaging elements for acquiring an image signal for the right eye and an image signal for the left eye to support three-dimensional (3D) display. The 3D display allows the operator **11131** to more accurately know the depth of a living tissue in a surgical site. Note that, in a case where the imaging unit **11402** includes multi-plate type solid-state imaging elements, a plurality of the lens units **11401** may be provided corresponding to the individual solid-state imaging elements.

(271) In addition, the imaging unit **11402** may not necessarily be provided in the camera head **11102**. For example, the imaging unit **11402** may be provided inside the lens barrel **11101** immediately behind the objective lens.

(272) The drive unit **11403**, which includes an actuator, causes the zoom lens and the focus lens in the lens unit **11401** to move by a predetermined distance along the optical axis under the control of the camera head control unit **11405**. As a result, the magnification and focus of an image captured by the imaging unit **11402** may be appropriately adjusted.

(273) The communication unit **11404** includes a communication device for sending and receiving various types of information to and from the CCU **11201**. The communication unit **11404** sends the image signal obtained from the imaging unit **11402** and treated as RAW data to the CCU **11201** via

the transmission cable **11400**.

(274) Furthermore, the communication unit **11404** receives control signals for controlling driving of the camera head **11102** from the CCU **11201** and supplies the control signals to the camera head control unit **11405**. Such control signals include, for example, information regarding imaging conditions such as information indicating that a frame rate of captured images is specified, information indicating that an exposure value applied during imaging is specified, and information indicating that magnification and focus of a captured image are specified.

(275) Note that the above-mentioned imaging conditions including frame rate, exposure value, magnification, and focus may be specified by the user as appropriate, or may be automatically set by the control unit **11413** in the CCU **11201** on the basis of the acquired image signals. In the latter case, the endoscope **11100** should include the so-called auto exposure (AE) function, auto focus (AF) function, and auto white balance (AWB) function.

(276) The camera head control unit **11405** controls driving of the camera head **11102** on the basis of control signals received from the CCU **11201** via the communication unit **11404**.

(277) The communication unit **11411** includes a communication device for sending and receiving various types of information to and from the camera head **11102**. The communication unit **11411** receives an image signal sent from the camera head **11102** via the transmission cable **11400**.

(278) Furthermore, the communication unit **11411** sends a control signal for controlling driving of the camera head **11102** to the camera head **11102**. Image signals and control signals can be transmitted by electric communication, optical communication, or the like.

(279) The image processing unit **11412** performs various types of image processing on an image signal, which is RAW data sent from the camera head **11102**.

(280) The control unit **11413** performs various types of control related to imaging of a surgical site or the like by the endoscope **11100** and display of a captured image obtained by imaging the surgical site or the like. For example, the control unit **11413** generates a control signal for controlling driving of the camera head **11102**.

(281) In addition, the control unit **11413** causes the display device **11202** to display a captured image showing the surgical site or the like, on the basis of the image signal that has undergone image processing by the image processing unit **11412**. During the process, the control unit **11413** may recognize various objects in the captured image using various image recognition techniques. For example, the control unit **11413** can recognize the surgical tool **11110** such as forceps, a specific site in a living body, bleeding, a mist produced when the energy treatment tool **11112** is used, and the like, by detecting edge shapes, colors, and the like of objects included in the captured image. When causing the display device **11202** to display the captured image, the control unit **11413** may use the recognition result to cause various types of surgery assisting information to be superimposed on the image of the surgical site. With the surgery assisting information superimposed on the display and presented to the operator **11131**, the operator **11131** bears less burden and can reliably proceed with the surgery.

(282) The transmission cable **11400** connecting the camera head **11102** and the CCU **11201** is an electric signal cable supporting electric signal communication, an optical fiber supporting optical communication, or a composite cable thereof.

(283) In the example illustrated here, the transmission cable **11400** is used for wired communication; however, communication between the camera head **11102** and the CCU **11201** may be carried out wirelessly.

(284) The foregoing has described an example of the endoscopic surgery system to which the technology according to the present disclosure may be applied. The technology according to the present disclosure may be applied to the imaging unit **11402** provided in the camera head **11102** of the endoscope **11100** among the above-described components. By applying the technology according to the present disclosure to the imaging unit **11402**, the imaging unit **11402** can be made smaller in size or have higher definition and therefore, the endoscope **11100** being small-sized or

having high definition can be provided.

Other Embodiments

(285) Note that the effects described herein are examples only and are not restrictive, and other effects may be provided.

(286) Furthermore, the present technology may have the following configurations.

(287) (1)

(288) A solid-state imaging element including: a first semiconductor substrate that includes a floating diffusion that temporarily holds an electric signal output from a photoelectric conversion element; and a second semiconductor substrate that faces the first semiconductor substrate, in which the second semiconductor substrate includes a first transistor disposed on a side facing the first semiconductor substrate, the first transistor including: a channel extending along a thickness direction of the second semiconductor substrate; and a multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the channel, and the multi-gate of the first transistor is connected to the floating diffusion.

(2)

(289) The solid-state imaging element according to (1), further including: a contact that connects a surface of the multi-gate and a surface of the floating diffusion, the surface of the multi-gate facing the surface of the floating diffusion.

(3)

(290) The solid-state imaging element according to (1) or (2), in which the second semiconductor substrate includes a second transistor disposed on a side facing the first semiconductor substrate, the second transistor including a source region, and the multi-gate of the first transistor is connected to the source region of the second transistor.

(4)

(291) The solid-state imaging element according to any one of (1) to (3), in which the second semiconductor substrate includes: a source region that extends from one surface side of the second semiconductor substrate and reaches another surface side of the second semiconductor substrate; and a drain region that extends from one surface side of the second semiconductor substrate and reaches another surface side of the second semiconductor substrate, and the source region is connected to a signal line that transmits the electric signal from a surface side of the second semiconductor substrate, the surface side being opposite to a surface facing the first semiconductor substrate, and the drain region is connected to a power supply potential from a surface side of the second semiconductor substrate, the surface side being opposite to the surface facing the first semiconductor substrate.

(5)

(292) The solid-state imaging element according to (1) or (2), in which the second semiconductor substrate includes a second transistor disposed on a side facing the first semiconductor substrate, the second transistor including: a channel extending along a thickness direction of the second semiconductor substrate; and a multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the channel, and the multi-gate of the second transistor is connected to a signal line that transmits the electric signal from a surface side of the second semiconductor substrate, the surface side being opposite to a surface facing the first semiconductor substrate.

(6)

(293) The solid-state imaging element according to (5), in which the first transistor is an amplification transistor that amplifies the electric signal that is output from the photoelectric conversion element, and the second transistor is a reset transistor that resets a potential of the multi-gate of the amplification transistor to a power supply potential.

(7)

(294) The solid-state imaging element according to (6), in which the second semiconductor

substrate includes a selection transistor that selects whether or not to transmit the electric signal amplified in the amplification transistor to the signal line.

(8)

(295) The solid-state imaging element according to (6), further including: a third semiconductor substrate disposed on a side opposite to the first semiconductor substrate, the third semiconductor substrate facing the second semiconductor substrate, in which the third semiconductor substrate includes a selection transistor that selects whether or not to transmit the electric signal amplified in the amplification transistor to the signal line.

(9)

(296) The solid-state imaging element according to any one of (6) to (8), in which the first semiconductor substrate includes a transfer transistor that transfers the electric signal to the amplification transistor, the electric signal being output from the photoelectric conversion element.

(10)

(297) A video recording apparatus including: a solid-state imaging element; an optical system that takes incident light from a subject to form an image on an imaging area of the solid-state imaging element; and a signal processing circuit that performs processing on an output signal from the solid-state imaging element, in which the solid-state imaging element includes: a first semiconductor substrate that includes a floating diffusion that temporarily holds an electric signal output from a photoelectric conversion element; and a second semiconductor substrate that faces the first semiconductor substrate, the second semiconductor substrate includes a first transistor disposed on a side facing the first semiconductor substrate, the first transistor including: a channel extending along a thickness direction of the second semiconductor substrate; and a multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the channel, and the multi-gate of the first transistor is connected to the floating diffusion.

REFERENCE SIGNS LIST

(298) **100** Solid-state imaging element **200, 300, 400** Stacked body **201, 301, 401** Substrate **203** Photoelectric conversion element **204** HAD **220** Transfer transistor **221** FD **221c, 312c, 322c, 323c, 331c** Contact **310** Amplification transistor **311, 321, 331** Source region **312, 322, 332** Drain region **313** Gate electrode **313d** Wiring **320** Reset transistor **330** Selection transistor

Claims

1. A solid-state imaging element, comprising: a first semiconductor substrate that includes a floating diffusion that temporarily holds an electric signal output from a photoelectric conversion element; and a second semiconductor substrate that faces the first semiconductor substrate, wherein the second semiconductor substrate includes: a first transistor disposed on a side facing the first semiconductor substrate, the first transistor including: a first channel extending along a thickness direction of the second semiconductor substrate; and a first multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the first channel, wherein a first contact is connected to the first multi-gate at the side facing the first semiconductor substrate; and a second transistor disposed on the side facing the first semiconductor substrate, the second transistor including: a second channel extending along the thickness direction of the second semiconductor substrate; and a second multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the second channel, wherein a second contact is connected to the second multi-gate at a side opposite the side facing the first semiconductor substrate.

2. The solid-state imaging element according to claim 1, wherein the first contact connects a surface of the first multi-gate and a surface of the floating diffusion, the surface of the first multi-gate facing the surface of the floating diffusion.

3. The solid-state imaging element according to claim 1, wherein the first multi-gate of the first

transistor is connected to a source region of the second transistor.

4. The solid-state imaging element according to claim 1, wherein the second semiconductor substrate includes: a source region of the first transistor that extends from one surface side of the second semiconductor substrate and reaches another surface side of the second semiconductor substrate; and a drain region of the first transistor that extends from one surface side of the second semiconductor substrate and reaches another surface side of the second semiconductor substrate, and the source region of the first transistor is connected to a signal line that transmits the electric signal from a surface side of the second semiconductor substrate, the surface side being opposite to a surface facing the first semiconductor substrate, and the drain region of the first transistor is connected to a power supply potential from a surface side of the second semiconductor substrate, the surface side being opposite to the surface facing the first semiconductor substrate.
5. The solid-state imaging element according to claim 1, wherein that the second contact transmits an electric signal from a surface side of the second semiconductor substrate, the surface side being opposite to a surface facing the first semiconductor substrate.
6. The solid-state imaging element according to claim 5, wherein the first transistor is an amplification transistor that amplifies the electric signal that is output from the photoelectric conversion element, and the second transistor is a reset transistor that resets a potential of the multi-gate of the amplification transistor to a power supply potential.
7. The solid-state imaging element according to claim 6, wherein the second semiconductor substrate includes a selection transistor that selects whether or not to transmit the electric signal amplified in the amplification transistor to a signal line.
8. The solid-state imaging element according to claim 6, further comprising: a third semiconductor substrate disposed on a side opposite to the first semiconductor substrate, the third semiconductor substrate facing the second semiconductor substrate, wherein the third semiconductor substrate includes a selection transistor that selects whether or not to transmit the electric signal amplified in the amplification transistor to a signal line.
9. The solid-state imaging element according to claim 6, wherein the first semiconductor substrate includes a transfer transistor that transfers the electric signal to the amplification transistor, the electric signal being output from the photoelectric conversion element.
10. The solid-state imaging element according to claim 1, wherein the first channel includes a part of the second semiconductor substrate.
11. The solid-state imaging element according to claim 10, wherein a first gate insulating film is interposed between the first channel and the first multi-gate.
12. The solid-state imaging element according to claim 11, wherein the first multi-gate is connected on three sides to the first channel via the first gate insulating film.
13. The solid-state imaging element according to claim 10, wherein the second channel includes a part of the second semiconductor substrate.
14. The solid-state imaging element according to claim 13, wherein a second gate insulating film is interposed between the second channel and the second multi-gate.
15. The solid-state imaging element according to claim 14, wherein the second multi-gate is connected on three sides to the second channel via the second gate insulating film.
16. The solid-state imaging element according to claim 1, wherein the first multi-gate is connected to a source region of the second transistor.
17. The solid-state imaging element according to claim 16, wherein the first multi-gate extends towards the second transistor to form a wiring connecting the first multi-gate and the source region of the second transistor.
18. The solid-state imaging element according to claim 1, wherein the first transistor is configured as a tri-gate transistor.
19. The solid-state imaging element according to claim 18, wherein the second transistor is configured as a tri-gate transistor.

20. A video recording apparatus comprising: a solid-state imaging element; an optical system that takes incident light from a subject to form an image on an imaging area of the solid-state imaging element; and a signal processing circuit that performs processing on an output signal from the solid-state imaging element, wherein the solid-state imaging element includes: a first semiconductor substrate that includes a floating diffusion that temporarily holds an electric signal output from a photoelectric conversion element; and a second semiconductor substrate that faces the first semiconductor substrate, the second semiconductor substrate includes a first transistor disposed on a side facing the first semiconductor substrate, the first transistor including: a first channel extending along a thickness direction of the second semiconductor substrate; and a first multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the first channel, wherein a first contact is connected to the first multi-gate at the side facing the first semiconductor substrate; and the second semiconductor substrate includes a second transistor disposed on the side facing the first semiconductor substrate, the second transistor including: a second channel extending along the thickness direction of the second semiconductor substrate; and a second multi-gate extending along the thickness direction of the second semiconductor substrate and sandwiching the second channel, wherein a second contact is connected to the second multi-gate at a side opposite the side facing the first semiconductor substrate.
